

# Modelling, simulation, and optimisation of agrivoltaic systems: a comprehensive review

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## HIGHLIGHTS

- Modelling and simulation of agrivoltaic systems are fundamental to predict crop and energy performance before installation and meet regulatory frameworks.
- Integrated modelling platforms for agrivoltaic systems must holistically consider the processes affecting crop growth and energy performances.
- Future research should focus on standardised approaches and collaborative validation for various agrivoltaic configurations.

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## ABSTRACT

Agrivoltaic systems combine food production and solar energy conversion on the same land, offering a dual-use approach to address land use concerns in renewable energy development. One of the main research and market challenges for agrivoltaic systems is the ability to predict food and energy yields prior to installation. The photovoltaic modules reduce solar irradiation on the ground, altering the energy balance at the ground and crop levels, affecting thus evapotranspiration and photosynthesis. The photovoltaic modules also influence local rain distribution and wind patterns, creating a microclimate that impacts both crop production and photovoltaic efficiency. The need to evaluate these effects and their impact on crop growth before installation is underscored by the recent implementation of new standards, guidelines, and regulations governing agrivoltaic systems in various regions. This study provides a critical review of existing research with a focus on the modelling, simulation, and optimisation of agrivoltaic systems. It highlights recent advancements in simulating and optimising the design of agrivoltaic systems through integrated simulations of shading, microclimates, electrical performance, and agricultural productivity. This study highlights the critical role of optimised light distribution in enhancing both crop yields and electricity production within agrivoltaic systems. However, the diversity of modelling approaches from the PV and agricultural sectors, coupled with the absence of standardised benchmarks, complicates the selection of appropriate models for specific systems and conditions. Future research should prioritise the development of standardised benchmarks to enable consistent comparisons across models, facilitating a better understanding of trade-offs between computational efficiency, interpretability, and accuracy. Collaborative efforts, publicly available datasets, and benchmarking initiatives are essential for validating models across diverse agrivoltaic configurations and regions.

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## 1. Introduction

In recent years, the concept of agrivoltaic (APV) systems, which combine agricultural crop production with photovoltaic (PV) solar energy conversion on the same land area, has gained considerable attention as a solution to secure management of water, energy, and food resources, also known as the Water–Energy–Food (WEF) nexus [1,2].

As APV systems gain momentum globally, policies and regulations are being developed to support and govern its implementation. For instance, Germany, Japan, and the United States have made significant progress in introducing specific restrictions, support programmes, and crop yield regulations. The policies in Germany [3], Japan [4], and the United States [5] reflect a growing recognition of the importance of APV systems in achieving renewable energy targets while ensuring optimal agricultural output. In Germany, an APV system must be designed to ensure that the simultaneous production of electricity from PV modules does not reduce agricultural yield by more than one third compared to the reference yield without PV system [6]. Regulations regarding yield reduction vary from country to country. In Japan, the yield reduction should not exceed 20 % compared to the reference yield [7]. Not all APV regulations focus exclusively on crop yield reduction compared to open-field conditions. Some regulations focus on light reduction. For example, the State of Massachusetts in the United States stipulates that the amount of light reaching plants must not fall below 50 % of full light, which is meant to ensure high crop yields [5]. In Catalonia, irradiation reduction under APV systems must not exceed 40 %, except for shade-tolerant crops such as kiwis and berries, where light reduction is permitted to reach 50 % [8]. Similarly, new regulations for APV systems are emerging in more countries and must be adhered to. These regulations often include criteria similar to those above mentioned to ensure that the simultaneous production of electricity from PV modules does not significantly reduce agricultural yield [6].

In this regulatory context, assessing crop yield performance or changes in irradiation distribution prior to installation is crucial. Integrated modelling, simulation, and optimisation of APV systems are fundamental to meeting policy requirements while maximising economic aspects [9]. These processes aim to analyse how the shading produced by PV modules impacts solar irradiation distribution on the ground, the microclimate beneath the PV modules, how this microclimate influences crop growth, and, in turn, how microclimate and crop growth affects PV production. The development and simulation of APV systems have advanced rapidly, integrating diverse modelling disciplines, including irradiance, crop growth, PV production, and microclimate dynamics. These models seek to provide comprehensive estimates of both crop yield and electricity production. However, despite these advancements, there is an urgent need for robust and standardised models that can reliably inform policy decisions and support practical implementation. At present, the APV modelling landscape lacks a unified framework, resulting in uncertainty about which models are more appropriate for specific contexts.

While several review papers have been published on APV systems, most focus on broader aspects of the technology, such as its potential for sustainable energy conversion and the integration of agricultural practices [9–16]. These reviews provide valuable insights into the overall functioning of APV systems but do not explore the modelling methodologies or their comparative effectiveness. Consequently, a significant gap exists in the literature regarding the complexities of APV system modelling, including accuracy, validation, regional applicability, and optimisation. The absence of standardised models or common benchmarks for evaluating and comparing APV systems complicates decision-making. Models differ widely in their scope, assumptions, and methodologies, making it difficult to assess their relative effectiveness. Additionally, there is no consensus on best practices for selecting the most appropriate model for specific APV scenarios, particularly regarding factors such as location, crop type, and environmental conditions. To address this gap, this review aims to consolidate existing APV

modelling efforts, highlighting key studies and their respective methodologies. By evaluating the accuracy, validation procedures, and regional applicability of these models, this work seeks to provide clearer guidance on the selection of APV models for simulation and optimisation. Furthermore, by synthesising insights from relevant studies, the review presents the current state-of-the-art in APV modelling, offering support for decision-making within the sector and advancing the WEF nexus. These insights are critical for researchers, policymakers, and industry professionals engaged in the APV sector. Ultimately, this review aims to contribute to the development of a standardised framework for future APV research, facilitating the large-scale integration of APV systems and supporting evidence-based policy development.

## 2. Methodology

This study employs a structured review process to explore and analyse the current state of research on APV systems, with a particular focus on modelling, simulation, and optimisation. The process is designed to comprehensively capture the latest findings. The methodology consists of three main stages: identification, screening, and inclusion as illustrated in Fig. 1.

Relevant research articles were sourced from established databases, including ScienceDirect, Web of Science, and Scopus. The search focused on literature using keywords such as agrivoltaic, agrophotovoltaic, and PV agriculture, combined with terms like modelling, simulation, and optimisation. Additionally, prior studies and their references were reviewed to expand the pool of relevant articles. Initially, 398 records were retrieved. Before progressing to the screening stage, 108 duplicate entries were identified and removed. During this phase, the remaining 290 articles were evaluated for inclusion based on the following criteria:

- Only articles written in English were considered.
- Articles needed to address APV modelling, simulation, or optimisation.
- Full-text versions of the articles had to be available for review.

Based on these criteria, 204 articles were excluded due to irrelevance, lack of a full-text version, or language incompatibility. After the screening process, 86 studies met the criteria for detailed analysis. To ensure a comprehensive and up-to-date understanding of APV research, 65 additional studies were manually added. These included newly published APV-focused articles and studies not specifically centred on APV but addressing key topics such as irradiance modelling, PV system modelling, crop modelling, microclimate studies, and optimisation platforms. This expanded the final dataset to a total of 151 studies.

### 2.1. Literature overview

Research on APV system modelling, simulation, and optimisation, along with related areas such as irradiance modelling, PV system modelling, crop modelling, and optimisation platforms, has experienced significant growth over the past decade, as illustrated in Fig. 2, which focuses exclusively on publications from the last 10 years. Between 2014 and 2016, the annual number of publications remained relatively low, with fewer than five articles published each year. From 2017 to 2019, a moderate increase was observed, with an average of 5 to 8 articles published annually. A notable surge occurred in 2021 and 2022, with the number of publications peaking at 23 in 2021. This rapid growth highlights an emerging trend in exploring innovative methods for modelling APV systems. In recent years (2022 to 2024), the publication rate has stabilised at a high level, ranging from 16 to 23 articles per year, demonstrating sustained interest in this research domain.

The key research themes within APV modelling and related fields are depicted in Fig. 3. The keyword co-occurrence network provides insights into dominant themes and important points in the APV literature. The analysis included all keywords occurring at least three times, with the

primary search terms excluded to avoid bias. The most used terms are “photovoltaics”, “photovoltaic”, “shading”, and “agriculture”. The clustering highlights the interdisciplinary nature of APV research, which blends agricultural science, energy engineering, and environmental sustainability. The connections between clusters demonstrate how technical concepts (e.g., energy and shading) intersect with agricultural applications.

Out of the 151 articles investigated, 129 were journal articles and conference proceedings. Among these, the four most common journals were *Applied Energy* (15 %), *Solar Energy* (10 %), *Renewable Energy* (5 %), and the *European Journal of Agronomy* (5 %), while articles from other journals accounted for 66 %, as shown in Fig. 4. *Applied Energy* demonstrates significant interest in APV modelling, being the journal with the highest number of articles on this topic. The *European Journal of Agronomy* is a popular choice for publishing articles with an agricultural focus, whereas the *Solar Energy* journal tends to emphasise the solar irradiance and PV aspects.

### 3. Agrivoltaic modelling, simulation, and optimisation

An analysis of the performance of APV systems before installation is crucial to optimise their techno-economic efficiency and ensure sustainable agricultural practices. There are many factors that must be considered when optimising and modelling APV systems to achieve the optimal trade-off between electricity production and agricultural yield or meeting current regulatory framework or standards [9,10,13]. Fig. 5 illustrates the microclimatic variations that must be considered in APV models. Shading from the PV system causes a reduction in irradiation reaching the ground and crops. Moreover, the shading induces gradients in local ambient, crop, and soil temperatures, and the PV system might also cause changes in wind speed and direction. Thus, an accurate APV model must combine irradiance, PV system, microclimate, and crop modelling, which are all distinct but closely related fields of research.

A comprehensive review of irradiance modelling, PV modelling platforms, microclimate modelling, and crop modelling is provided in Sections 3.1, 3.2, 3.3, and 3.4, respectively. Integrated simulation and

optimisation platforms streamline the design process by combining multiple modelling aspects to improve the overall efficiency and reliability of APV systems. Integrated simulation platforms are reviewed in Section 3.5.

#### 3.1. Irradiance modelling

Solar irradiance is the driving force behind both crop growth and PV electricity production. Its modelling and distribution at the crop and PV array levels are of paramount importance for assessing both quality and quantity. Crops utilise only part of the solar irradiance spectrum (i.e., broadband Global Horizontal Irradiance [GHI]) for photosynthesis, and it corresponds to the region comprised between 400 and 700 nm, known as Photosynthetically Active Radiation (PAR). Within the PAR region, most crops exhibit peaks of absorbance in the red and blue wavelengths. In contrast, PV modules generally harvest a broader portion of the GHI spectrum, with the exact proportion depending on the solar cell technology. Different technologies exhibit varying efficiencies across different wavelengths. Furthermore, while crops are typically cultivated on horizontal or slightly tilted surfaces, PV modules are generally installed at a tilt relative to the horizontal plane and oriented to optimise the harvesting of solar irradiance. To accurately assess the irradiance reaching tilted and oriented PV modules, a two-step modelling approach is typically employed to estimate the Global Tilted Irradiance (GTI) from the input GHI, i.e., GHI decomposition and transposition. In APV systems, PV modules introduce shading on the ground, thereby reducing the incident PAR and GHI. Additionally, as with conventional PV systems, APV modules also produce self-shading. This Section is divided into key subsections covering decomposition models (3.1.1), transposition models (3.1.2), shading modelling (3.1.3), and PAR modelling (3.1.4), followed by the validation of these models (3.1.5). This Section critically reviews strategies for accurately modelling irradiance distribution, with a particular focus on enhancing system performance in dual land-use applications.

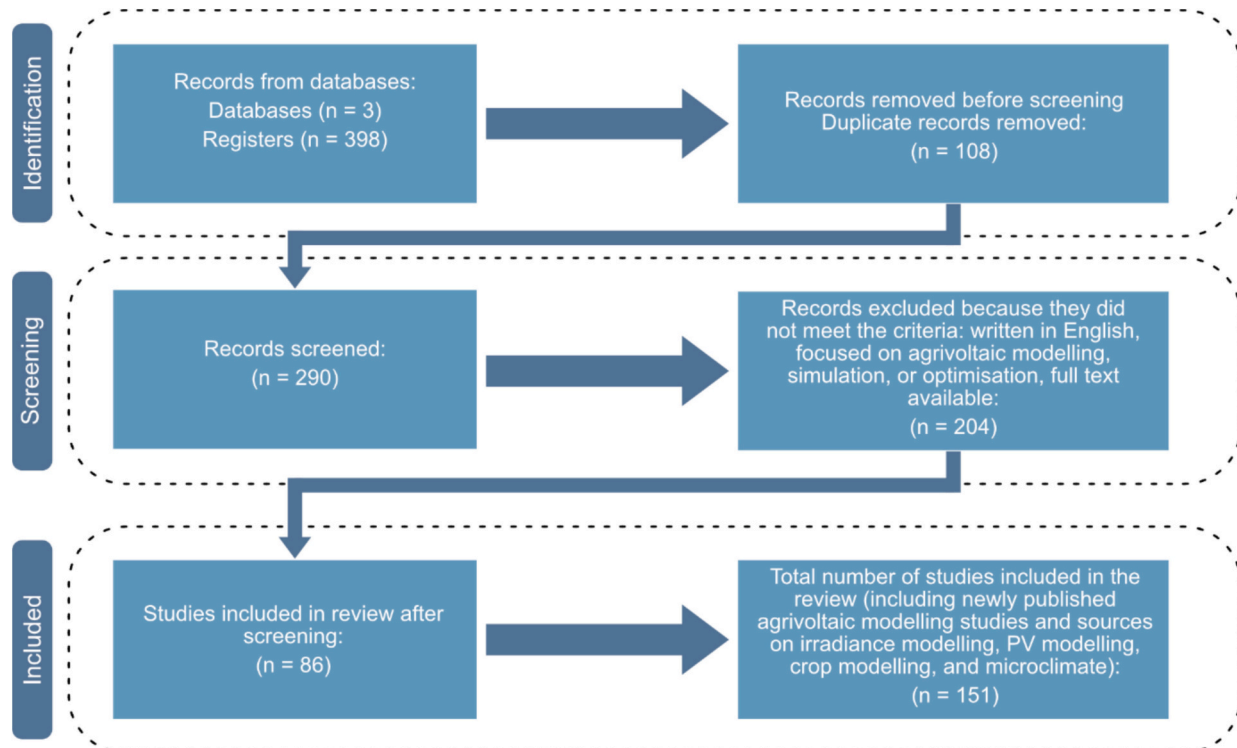


Fig. 1. Structured review process for APV research: identification, screening, and inclusion stages.

### 3.1.1. Decomposition models

Decomposition models, also referred to separation models of solar irradiance, are employed to separate Beam and Diffuse Horizontal Irradiance (BHI and DHI, respectively) from GHI. These models are often applied in the initial stages of a broader analysis when separate measurements of irradiance components are unavailable. They enable the estimation of BHI or Direct Normal Irradiance (DNI) from GHI measurements, typically using relationships between the clearness index and the diffuse fraction of solar irradiance. Like clear-sky models, decomposition models vary in complexity. One of the commonly used decomposition models was proposed by Liu and Jordan [17]. It was originally developed for daily and monthly time scales but was later widely adopted for estimating hourly DHI values. Over the years, numerous decomposition models have been developed to estimate the components of solar irradiance. For instance, Gueymard and Ruiz-Arias [18] conducted a benchmarking study of 140 decomposition models, using 1-min, high-quality GHI and DNI data collected from 54 research-grade stations across seven continents (Fig. 6).

The benchmark highlights that model accuracy varies depending on local conditions and input parameters. Models incorporating clear-sky estimates generally perform better. Among them, the model by Engerer [19] (ENERGER2) is a widely used decomposition model. It applies a logistic function-based approach with multiple predictors, including solar zenith angle, cloud enhancement effects, and clearness index trends, making it particularly effective in low-albedo environments where ground reflections are minimal. However, as with most empirical models, its performance may vary across different climates. In contrast, the model by Perez et al. [20] (PEREZ1) estimates DNI using an empirical approach that applies different coefficient sets based on atmospheric conditions. Instead of relying on a single equation, PEREZ1 categorises sky conditions (e.g., clear, intermediate, or overcast) based on clearness index and air mass and then applies condition-specific empirical relationships. This method allows it to better capture variations in direct irradiance across different atmospheric conditions. PEREZ1 is widely used in temperate regions, where atmospheric variability is moderate, and has been shown to perform well across various sky conditions. However, it can be less accurate in high-albedo environments, such as snowy or highly reflective surfaces where additional correction factors may be needed.

In a related development, Ruiz-Arias and Gueymard [21] introduced the GISPLIT model, which dynamically classifies sky conditions to enhance the decomposition of GHI into its direct and diffuse components. The model's standard version (G1) uses a single fit approach without climate stratification and is trained on data from all sites.

Versions that incorporate climate stratification and machine learning techniques show marginal improvements. Notably, version 3 (G3), which employs extreme gradient boosting, achieves the lowest DNI prediction error. G3 reduces the Root Mean Square Error (RMSE) from  $15.9 \text{ W m}^{-2}$  (G1) to  $15.3 \text{ W m}^{-2}$ , demonstrating a small but meaningful enhancement in accuracy.

APV modelling studies have primarily focused on integrating existing models, such as those by Collares-Pereira [22], Gu [23], and Orgill-Hollands [24–26], into the modelling framework, rather than benchmarking existing models or developing novel GHI decomposition models. The key outcomes, challenges, and future directions for decomposition modelling in APV systems are summarised in Table 1.

### 3.1.2. Transposition models

Transposition models are commonly used to calculate the incident irradiance on a tilted plane and can be divided into two main categories: isotropic and anisotropic models. Isotropic models, such as those developed by Liu and Jordan [27] or Badescu [28], assume that incoming diffuse irradiance is uniform in all directions, with no obstructions or directional effects. In contrast, anisotropic models, such as the Hay model [29], separate diffuse irradiance into two components: an isotropic contribution and a circumsolar contribution. More advanced anisotropic models, such as the Perez model [30], add a third diffuse component by introducing a horizon band, making them more sophisticated in accounting for directional effects.

Yang et al. [31] evaluated ten transposition models in Singapore, demonstrating that anisotropic models, such as the Hay and Skartveit-Olseth models, outperformed isotropic ones. For tilt angles below  $20^\circ$ , the RMSE improvements were lower ( $\sim 0.4\%$ ) compared to the measured irradiance under all-sky conditions. In contrast, isotropic models returned higher RMSE values ( $\sim 2\%$ ) as the tilt angles were adjusted from  $6.1^\circ$  to  $18.3^\circ$ , underscoring their limitations compared to anisotropic methods. Similarly, Nassar et al. [32] investigated the performance of 24 transposition models across various Palestinian cities, focusing on systems with a  $30^\circ$  southward tilt. Their findings revealed that no single model consistently outperformed others across all tested cities, highlighting the importance of selecting models based on specific local conditions. The transposition models most commonly used in APV modelling studies include the Hottel and Woertz [33], Liu and Jordan [34], Hay–Davies [34], and Perez models [23,24,26,35].

Anisotropic models excel in their ability to distribute the direct and diffuse irradiance components onto tilted surfaces under varying atmospheric conditions [34]. This structural advantage enables anisotropic models to consistently predict higher levels of irradiance

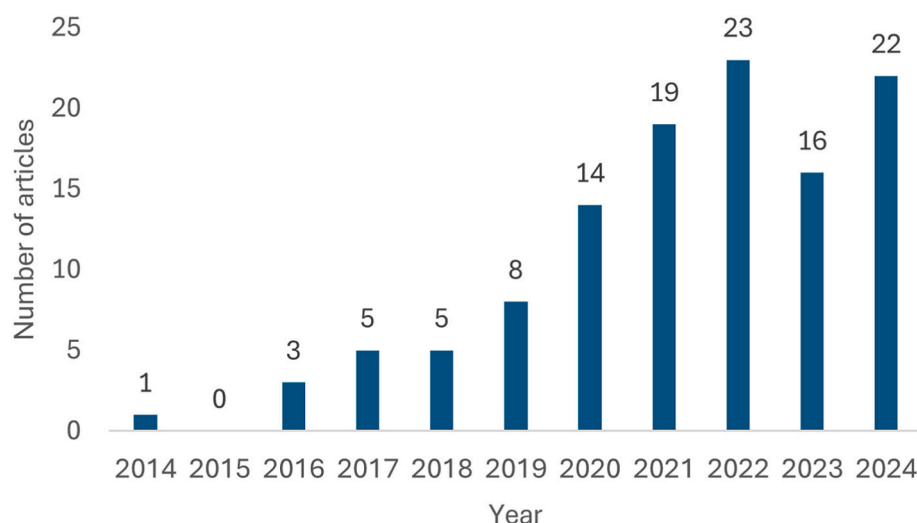


Fig. 2. Annual growth in publications on APV system modelling, simulation, optimisation, and related research, highlighting key trends over the past decade.



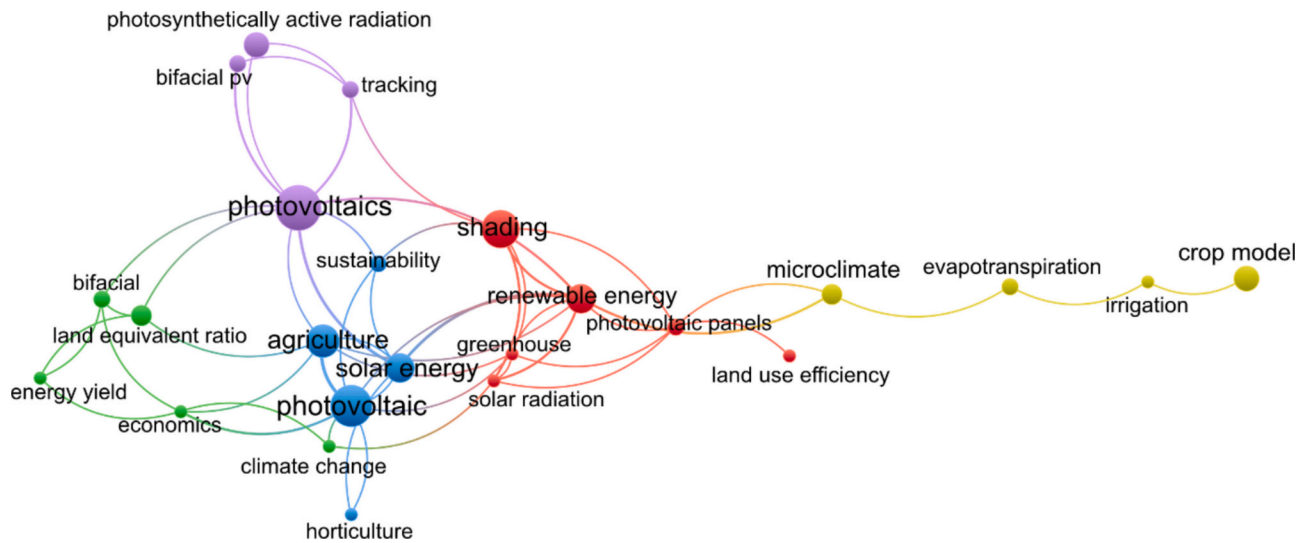


Fig. 3. Keyword co-occurrence network for APV research and related literature, showcasing dominant themes and methodological connections in modelling, simulation, and optimisation research.

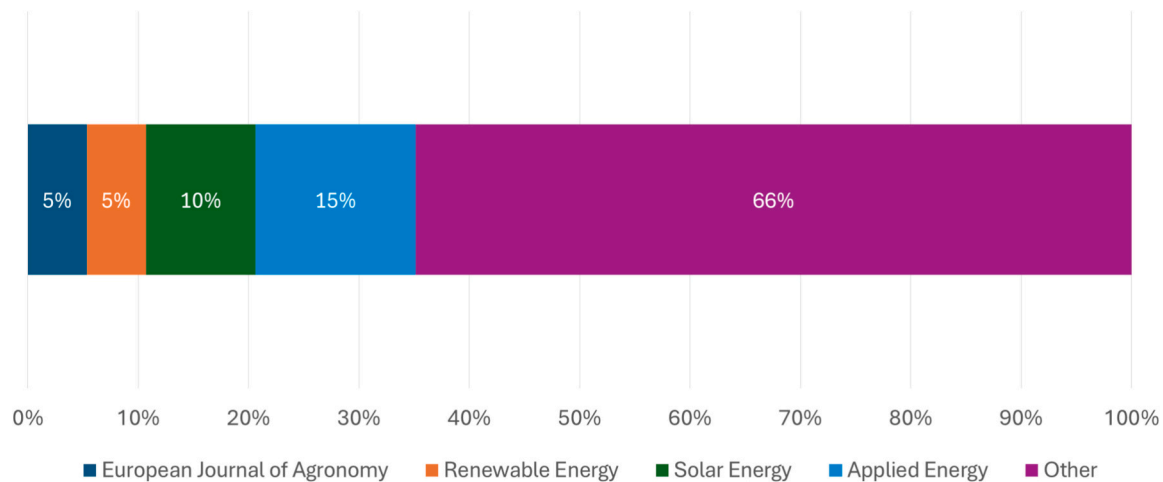


Fig. 4. Distribution of journal and conference proceeding publications on APV systems and related research included in this review.

compared to isotropic models, making them particularly effective when used with solar trackers. Furthermore, this improved capture of irradiance is evident not only under direct sunlight conditions but also in scenarios where direct irradiance is absent, with anisotropic models maintaining higher diffuse irradiance levels than isotropic models. However, despite their advantages, these models require further adaptation to accommodate the unique characteristics of APV installations, such as bifacial and vertical modules [36,37].

Robledo et al. [37] highlighted the challenges and critical considerations of applying transposition models to vertically mounted bifacial PV modules. The uncertainty associated with transposition models is generally higher for vertical surfaces. One reason for this increased uncertainty is that, for vertical surfaces, a substantial portion of the total diffuse radiation (sky + reflected) originates from ground-reflected irradiance, unlike in the case of low-tilt surfaces. To date, most transposition models have not adequately accounted for this ground-reflected diffuse irradiance, with only a few exceptions. Fig. 7 illustrates the contribution of reflected isotropic diffuse irradiance as a fraction of the total isotropic diffuse incident irradiance for a vertical APV configuration.

This suggests a potential shift towards favouring models that can more accurately capture complex irradiance interactions on vertical

APV installations. Such considerations are crucial for future APV projects and for enhancing the accuracy of energy yield predictions in these innovative systems. The key outcomes, challenges, and future directions for transposition modelling in APV systems are summarised in Table 2.

### 3.1.3. Shading modelling

Shading is a well-recognised issue in the PV industry, as it poses a significant challenge to the performance of PV systems, particularly due to self-shading between rows. Shading modelling techniques are typically based on geometric projections [38]. In the context of APV systems, the impact of shading extends beyond the PV modules themselves and affects the performance of crops. Shading reduces both the direct and diffuse irradiance available to the crops. The use of bifacial modules in APV systems further complicates shading analysis, as it makes estimating the reflected irradiance reaching the rear side of the PV modules more challenging.

Traditionally, calculating irradiance on a tilted surface relies on a 2D view-factor approach, which is based on several assumptions [22]. These include the infinitely long array, isotropic and homogeneous light reflection from the ground, and the visibility of the ground from the array's slope to the infinite horizon. To address these limitations, two common techniques are used to account for shading in APV system

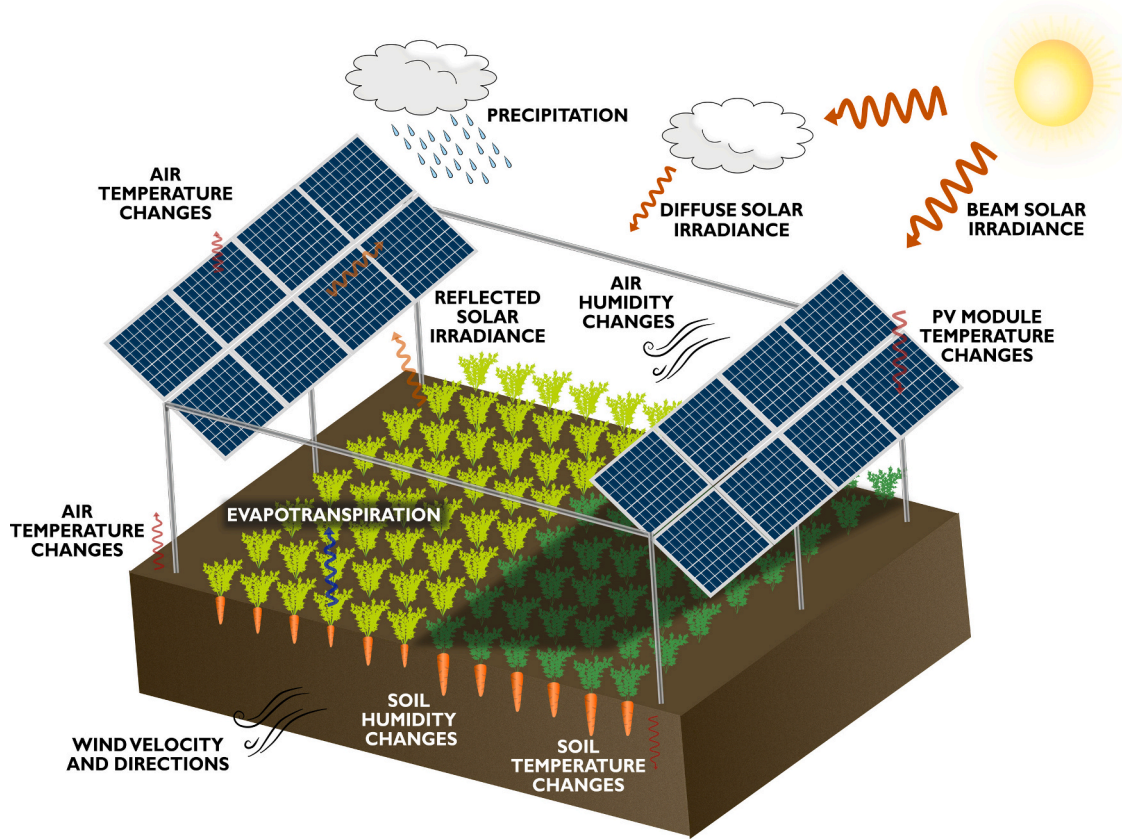


Fig. 5. A visual representation of the microclimate variations within an APV system.

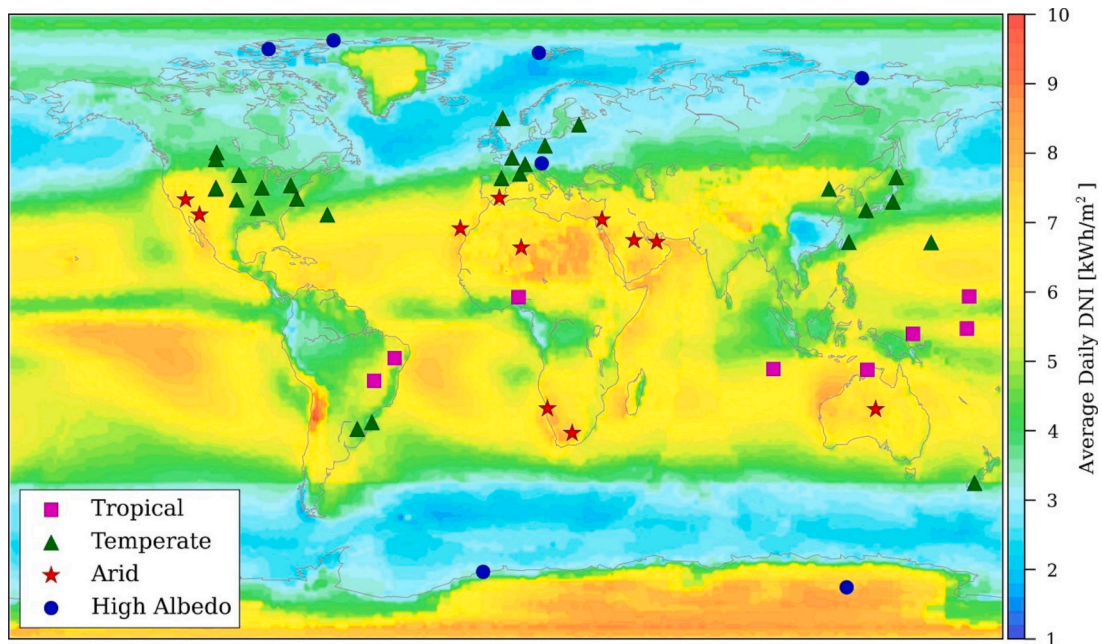


Fig. 6. Gueymard and Ruiz-Arias [18] benchmarking study of 140 decomposition models utilised data from 54 stations, whose geographic distribution is illustrated on a map overlaying the average daily DNI solar resource, as obtained from the NASA-SSE database. The stations are categorised into four climate zones represented by different symbols.

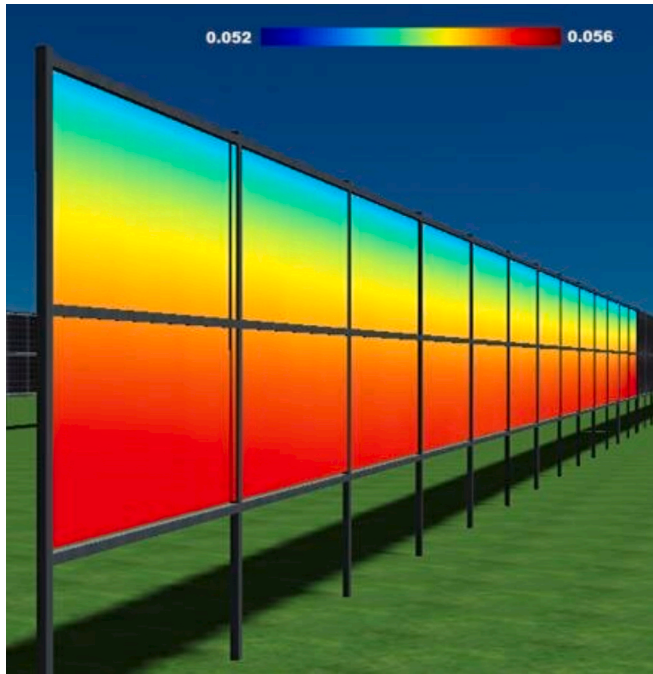
modelling: projection and view-factors (2D or 3D) [22–26,33,34,39–44], and ray-tracing [2,35,45–47]. These methods aim to enhance modelling accuracy by considering the heterogeneous nature of ground-reflected irradiance in APV systems, shading from

nearby objects, and row-to-row shading. Each method has its strengths and limitations. Projection or view-factor modelling is effective for estimating the shadows cast by the PV modules without being overly computationally intensive. However, accurately modelling complex

**Table 1**

Key outcomes, challenges, and future directions for decomposition modelling in APV systems.

| Key Outcomes                                                                                                                                                                                                                                                               | Challenges                                                                                                                                                                                                                                                                                                                                                                                                                       | Future Directions                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• ENGERER2 model excels in low-albedo environments.</li> <li>• PEREZ1 consistent in temperate climates.</li> <li>• Taking climate classification into account does not significantly enhance accuracy over all climates.</li> </ul> | <ul style="list-style-type: none"> <li>• Model accuracy varies with local conditions and model inputs.</li> <li>• No model reliably predicts under cloud enhancements and high-albedo conditions.</li> <li>• General model performance varies across climate zones, highlighting the need for regional adaptations.</li> <li>• Recent models show limited improvements, indicating potential issues with overfitting.</li> </ul> | <ul style="list-style-type: none"> <li>• Introducing models such as GISPLIT to enhance GHI decomposition by classifying sky conditions and utilising machine learning techniques.</li> <li>• Models should focus on refining climate-specific adaptations and integrate advanced data analytics to better capture regional variations and improve accuracy.</li> </ul> |



**Fig. 7.** Contribution of reflected isotropic diffuse irradiance to total isotropic diffuse incident irradiance in a vertical APV configuration [37].

crop geometries using geometric projections can be challenging and may reduce the precision of the results [38]. Torrente et al. [48] applied geometric projection to simulate shading effects within APV greenhouses. By simplifying greenhouse structures and PV panel layouts into basic geometric elements, their model calculated irradiation distribution, including both direct and diffuse components, while accounting for obstructions via sky view-factors. The study found that the predicted radiation reductions were slightly lower than the rooftop shading percentages. This discrepancy was attributed to the contribution of unshaded irradiance from greenhouse walls and the orientation of the greenhouse, which often caused PV panel shadows to fall outside its boundaries. These effects were more pronounced at higher shading percentages.

In contrast, ray-tracing has become a widely used technique for determining the fraction of light that reaches both PV modules and crops. Riedel-Lyngskær [49] benchmarked eight state-of-the-art view-factor and ray-tracing methods for rear irradiance estimations in

**Table 2**

Key outcomes, challenges, and future directions for transposition modelling in APV systems.

| Key Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Challenges                                                                                                                                                                                                                                                                                             | Future directions                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Anisotropic models such as Hay and Skartveit &amp; Olseth provide higher irradiance predictions with lower relative RMSE compared to isotropic models.</li> <li>• Anisotropic models are better suited for environments with complex light interactions. They can also effectively separate different components of diffuse irradiance.</li> <li>• Commonly used models in APV studies include Hottel and Woertz, Liu and Jordan, Hay–Davies, and Perez models.</li> <li>• Minimal variation between transposition models for tilt angles <math>&lt;20^\circ</math>.</li> </ul> | <ul style="list-style-type: none"> <li>• Isotropic models are limited to uniform irradiance scenarios and do not account for directional effects.</li> <li>• Anisotropic models require adjustments for bifacial and vertical modules.</li> <li>• Performance varies by region and climate.</li> </ul> | <ul style="list-style-type: none"> <li>• Develop hybrid models combining both isotropic and anisotropic strengths.</li> <li>• Integrate real-time data for dynamic adjustments.</li> <li>• Enhance model reliability with extensive calibration and validation.</li> </ul> |

Roskilde, Denmark. In fixed-tilted PV systems, the view-factor approach achieved accuracy within  $\pm 1\%$  of the measured monthly bifacial gain. However, it is less accurate for single-axis tracking systems, within  $\sim 2\%$ , while the ray-tracing approach remained within  $1\%$  of the measured bifacial gain. Kang et al. [50] analysed the front and rear irradiance in an elevated APV system and found that ray-tracing significantly reduced both Mean Bias Error (MBE) and RMSE compared to 2D view-factors. The front irradiance RMSE decreased by approximately  $2\%$ , while the rear irradiance RMSE decreased by approximately  $16\%$  when ray-tracing was used instead of 2D view-factors. Although ray-tracing models can handle complex geometries, they are computationally intensive, making them less practical for optimisation and comparisons of different system designs. However, ray-tracing models can easily account for reflections and refractions by defining optical properties for each geometry [35]. These reflections must be considered separately by taking them into account in either projection or shading models. Campana et al. [31] combined 2D view-factors with Monte Carlo analysis to estimate the impact of PV module reflections on crops. Jamil et al. [51] integrated various simulation engines, including ray-tracing software, such as DAYSIM and Radiance, along with thermal modelling tools, to create interactive visualisations of light distribution in vertical APV systems using the Rhinoceros 3D modelling environment. Building on the work of Ernst et al. [52], their study demonstrated significant improvements in precision for complex systems using ray-tracing methods. The researchers systematically introduced various geometric and optical complexities into the ray-tracing simulations, showcasing its robustness in accurately modelling rear-side irradiance. While ray-tracing and view-factor models produce similar results for front-side irradiance, where the geometric complexity is minimal, the inclusion of structural elements, such as module frames and torque tubes, has a pronounced impact on rear irradiance predictions. Specifically, the ray-tracing model accounts for shading effects and inter-reflections, leading to more accurate rear-side irradiance simulations. Both models exhibit comparable biases for front-side irradiance, with ray-tracing showing a bias of  $-1.8\%$  and the view-factor model a bias of  $-2.0\%$ . However, for rear-side irradiance, the ray-tracing model demonstrates a minimal bias of  $-0.1\%$ , whereas the view-factor model significantly underestimates irradiance with a bias of  $-8.4\%$ . Expanding on the advantages of ray-tracing, Mouhib et al. [53] used ray-tracing to simulate APV systems



combined with olive tree orchards. In their study, olive trees were represented as cylindrical volumes to accurately model their shading effects on underlying crops and bifacial PV panels (Fig. 8).

In their simulations, the number of tree rows was varied from 8 to 18 to assess the impact of field size on central-row irradiance. The results showed that increasing the number of rows beyond eight had minimal effects on irradiance, with only a 0.27 % variation observed for the central trees. The key outcomes, challenges, and future directions for transposition modelling in APV systems are summarised in Table 3.

### 3.1.4. Photosynthetically active radiation

The spectral range of solar radiation between 400 and 700 nm, known as PAR, is essential for photosynthesis in plants. Photosynthesis is the process by which plants and other organisms convert solar energy into chemical energy [55]. It is typically expressed either as Photosynthetic Photon Flux Density (PPFD,  $\mu\text{mol photons m}^{-2} \text{s}^{-1}$ ), since photosynthesis is a quantum process, or as photosynthetic radiant flux density (PAR,  $\text{W m}^{-2}$ ), which is more suitable for energy balance studies. In cases where direct PAR measurements are unavailable, empirical models are commonly used to estimate PAR based on relationships with broadband irradiance parameters, often using GHI as the primary input. For example, the model proposed by Varo-Martínez et al. [56] utilised the linear equation from Mariscal et al. [57], which estimates total PAR by multiplying GHI by a factor of 0.45. Similarly, Jin et al. [58] followed an approach that translates GHI to PAR, using a factor representing the PAR to GHI ratio, although they did not specify a fixed value. Hanrieder et al. [59] applied the Vindel et al. [60] model, a linear equation with two adjustable factors that vary by location and therefore require site-specific calibration to derive PAR from GHI values. However, such models are often location-specific, necessitating adjustments for different regions, and limiting their general applicability [61]. Artificial Neural Networks (ANNs) have emerged as a powerful tool for PAR modelling, thanks to their ability to capture complex nonlinear relationships between input variables and PAR. García-Rodríguez et al. [62] employed ANNs alongside multilinear regression to develop PAR models under varying sky conditions. However, the improvement in accuracy using ANNs over multilinear regression was not significant, as both methods achieved  $R^2$  greater than 0.97 under all sky conditions. Additionally, the complexity of ANNs and their reliance on extensive training data limit their widespread use. Like decomposition models, the accuracy of PAR estimation models decreases under cloudy conditions due to the high uncertainty associated with cloud optical properties [63].

Understanding the photosynthetic response of different crops is a complex task, as their behaviour varies under different environmental conditions. Emmel et al. [64] examined how the structure of the canopy in six major arable crops affects photosynthesis under diffuse light conditions. The study found that canopy photosynthesis is enhanced under diffuse light due to canopy's structure. This indicates that the architecture of the plant canopy plays a significant role in photosynthetic efficiency, particularly under diffuse light conditions. As a result, it is important to separate PAR into its direct and diffuse components to

**Table 3**

Key outcomes, challenges, and future directions for shading modelling in APV systems.

| Key Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Challenges                                                                                                                                                                                                                                                                                              | Future Directions                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Shading impacts PV system performance, especially self-shading.</li> <li>Shading in APV systems affects both PV modules and crops.</li> <li>Accurate assessment requires considering shading impacts on PV modules and plants.</li> <li>Shading reduces direct and diffuse irradiance to crops.</li> <li>Bifacial modules complicate irradiance estimation.</li> <li>Traditional 2D view-factor approach has assumptions like infinite array length and isotropic ground reflection.</li> <li>Two main shading modelling techniques: projection and view-factors, and ray-tracing.</li> <li>Ray-tracing is more accurate but computationally intensive.</li> </ul> | <ul style="list-style-type: none"> <li>Estimating reflected irradiance in bifacial modules.</li> <li>Geometric projection challenges in modelling complex crop geometries.</li> <li>Ray-tracing's high computational demands.</li> <li>Accounting for reflections and refractions in models.</li> </ul> | <ul style="list-style-type: none"> <li>Combine different techniques (e.g., 2D view-factors with Monte Carlo simulations) to improve accuracy.</li> <li>Develop methods to reduce computational demands of ray-tracing.</li> <li>Explore hybrid approaches to balance accuracy and computational efficiency.</li> </ul> |

accurately simulate PAR distribution within vegetation canopies [65]. Notably, the photosynthetic response is not limited to the visible spectrum (400–700 nm). Kukal and Irmak [66] observed that improving light distribution within crop canopies and expanding the range of light wavelengths beyond those traditionally associated with photosynthesis could have substantial implications for increasing crop yield and reducing the yield gap.

PAR decomposition models have a significant role in APV systems' modelling, simulation, and optimisation because crops undergo a complex dynamic combination of diffuse, reflected, and global irradiance, as shown in Fig. 9.

Ma Lu et al. [68] benchmarked seven decomposition models in Sweden by applying the Spitters relationship (Eq. 1) to each decomposition model to estimate diffuse PAR instead of DHI.

$$k_{\text{dPAR}} = \frac{\text{PAR}_{\text{diffuse}}}{\text{PAR}_{\text{total}}} = \frac{[1 + 0.3(1 - k_d^2)]k_d}{1 + (1 - k_d^2)\cos^2(90 - \beta)\cos^3(\beta)} \quad (1)$$

where  $k_d$  is the diffuse fraction and  $\beta$  is the solar elevation angle [ $^\circ$ ]. The denominator accounts for the correction of circumsolar irradiance, which arises due to the anisotropic nature of diffuse irradiance under clear sky conditions. Because of Mie scattering, diffuse irradiance is



**Fig. 8.** Representation of an APV system integrating olive trees, modelled using the bifacial\_radiance [54], a ray-tracing tool to capture incident irradiance at various points within the scene.



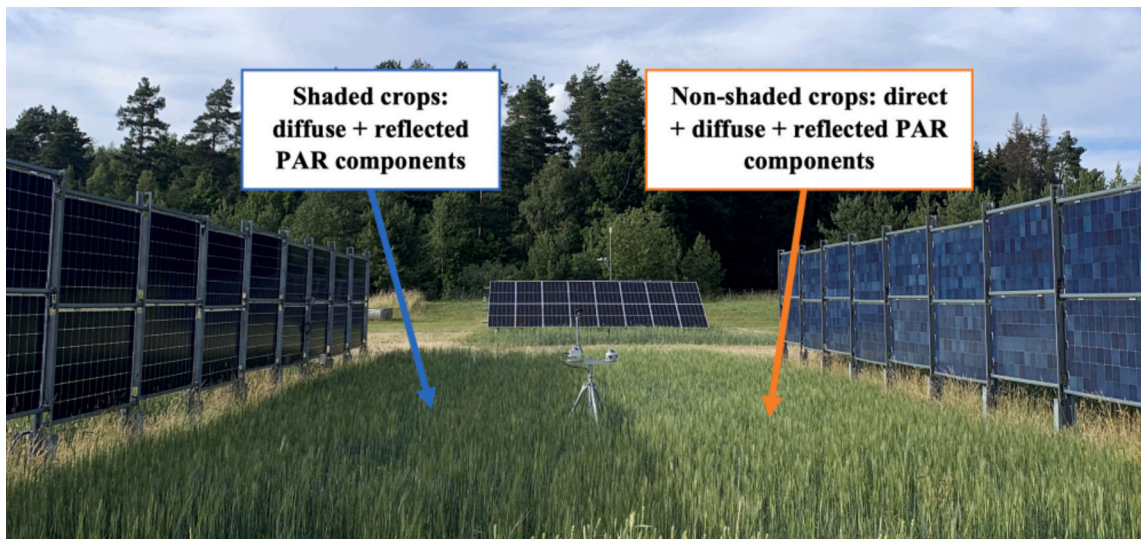


Fig. 9. PAR components received at the crop level in shaded and non-shaded conditions in a vertical APV system located in Sweden [67].

more intense in the direction of the sun. On a horizontal plane under clear skies, the circumsolar component is represented by  $\cos^2(90 - \beta)\cos^3(\beta)$ , which adjusts the diffuse flux [65]. To interpolate for intermediate sky conditions, the term  $1 + (1 - k_d^2)$  is introduced, further refining the diffuse fraction to better account for circumsolar irradiance. The numerator is based on empirical observations of diffuse irradiance within the PAR wavebands. Under clear skies, the scattered diffuse component in the PAR region is greater than in global irradiance due to the scattering properties of gases like nitrogen and oxygen [65]. It has been observed that the diffuse fraction in the PAR region is approximately 1.4 times that in global irradiance. However, to better represent photosynthetic activity across different wavebands, this factor is adjusted to 1.3, which is used in the numerator.

In the benchmarking study conducted by Ma Lu et al. [68], the model by Yang & Boland [69] (YANG2) was identified as the best-performing decomposition model based on normalised Mean Bias Error (nMBE), normalised Root Mean Squared Error (nRMSE), and  $R^2$ . The YANG2 model builds on previous logistic-function-based decomposition models, specifically ENGERER2 and the model developed by Starke et al. [70] (STARKE), and has been shown to outperform them in various climatic conditions. A key improvement in YANG2 over STARKE is its incorporation of satellite-derived diffuse fraction data. This addition enables YANG2 to smooth the decomposition process using only real-time input data, eliminating the need for future values, which is a limitation of STARKE. However, in most APV simulations, models are typically applied to historical data rather than used for forecasting, making this reliance on future values less relevant in practice. In contrast, the STARKE model also employs a logistic function-based approach but achieves smoothing by relying on future values of certain parameters. Specifically, STARKE uses a smoothed clearness index and the daily clearness index, both of which incorporate future values of the clearness index to create a more stable estimation. While this approach enhances accuracy, it restricts the model's application in real-time scenarios. Despite this limitation, STARKE has demonstrated high accuracy, outperforming ENGERER2 in multiple locations. When real-time satellite data are unavailable, STARKE remains a reliable alternative, particularly in environments where ground-based measurements, such as the clearness index, can be effectively used, even though the model requires future data for smoothing.

Building on the YANG2 model, Ma Lu et al. [67] proposed the CLY model to address the unique challenges of high-latitude regions. The CLY model retains the logistic functional form of YANG2 but introduces additional predictors, such as atmospheric optical thickness, Vapor

Pressure Deficit (VPD), and Aerosol Optical Depth (AOD), and surface albedo. These enhancements allow the model to better capture the scattering effects and variability of diffuse PAR, outperforming YANG2 and other models like STARKE, with  $R^2$  values exceeding 0.90 and lower normalised RMSE across studied sites. Despite its improved accuracy, the increased number of predictors raises challenges, particularly in terms of data availability. Satellite-derived inputs like AOD and surface albedo, while useful, can suffer from low spatial and temporal resolution, which may affect localised or small-scale applications. Additionally, the CLY model showed reduced performance under clear-sky conditions with low diffuse PAR values, a limitation that could impact its use during specific growing seasons. Addressing these issues, especially for early and late crop growth periods, and simplifying predictor requirements without sacrificing accuracy could be valuable directions for future work.

While existing irradiance models typically use the horizontal height of the upper canopy as the primary variable for estimating the total irradiance received by the crop [22,23,35,39], this approach does not fully capture the complexity of real-world conditions. The irradiance received by an individual leaf is influenced by numerous factors, including the spatial arrangement of the leaves, the degree of leaf overlap, and the position of the sun in the sky [39]. To develop more accurate irradiance models, additional factors affecting light distribution within the canopy must be considered. By accounting for the spatial arrangement of leaves and overlapping foliage, we can gain a better understanding of how crops receive irradiance at various growth stages [71]. Furthermore, incorporating the dynamic position of the sun throughout the day and its effects on light penetration will enhance the accuracy of our irradiance estimates. The key outcomes, challenges, and future directions for PAR modelling in APV systems are summarised in Table 4.

### 3.1.5. Validation of solar irradiance and photosynthetically active radiation models

Numerous studies have investigated models designed to estimate both the total irradiance reaching PV modules and the PAR at crop level in APV systems. Table 5 summarises the various types of solar irradiance models, including their modelled parameters, validation methods, and validation results. There are many models available for calculating Plane-Of-Array (POA) irradiance. These models typically simulate the different irradiance components, apply transposition approaches, and analyse shading caused by nearby obstacles. While these models have been extensively validated, most of them are focused on the incident

**Table 4**

Key outcomes, challenges, and future directions for PAR modelling in APV systems.

| Key Outcomes                                                                                                                                                                                                                                                                                                                                  | Challenges                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Future Directions                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Empirical models estimate PAR using GHI but are often location specific.</li> <li>ANNs have been used to model PAR, showing no significant accuracy improvement over multilinear regression.</li> <li>Decomposition of PAR into direct and diffuse can be done using Spitters relationship.</li> </ul> | <ul style="list-style-type: none"> <li>Empirical models require regional adjustment which limits their broader application.</li> <li>Accuracy under cloudy conditions remains a challenge due to uncertainty in cloud optical properties.</li> <li>ANNs, while powerful, require extensive data for training and are complex to implement.</li> <li>Traditional irradiance models often overlook the complexity of canopy structures and light distribution within them.</li> </ul> | <ul style="list-style-type: none"> <li>Models need to consider the dynamic and complex distribution of diffuse and reflected PAR in APV systems.</li> <li>Development of models that account for the spatial arrangement of leaves and overlapping foliage that affect light interception, shading, and photosynthetic activity.</li> </ul> |

irradiance on the PV modules. Many studies report models that accurately predict ground shading conditions, irradiance on PV modules, and PAR levels. For example, the models developed by Zainali et al. [44], Willockx et al. [72], and Zhang et al. [73] demonstrated high precision, with only minor deviations from actual measurements or commercial software comparison.

The studies either validate their models against established software or through direct field measurements to confirm the reliability under various APV configurations. Models specifically designed for decomposing global radiation into PAR components showed varying degrees of accuracy, with some models like those by Ma Lu et al. [68] initially showing weak results but improving significantly when tailored with local data. Models developed to map the spatial and temporal variations of irradiance within APV systems have been effective, as evidenced by Tahir et al. [33] and Santra et al. [79], who observed close alignment between model predictions and experimental data. However, there are still challenges such as simulating the diffuse and direct components of solar irradiance, especially under varying weather conditions, as noted by Wang et al. [75]. Additionally, the performance of some models is heavily dependent on local calibration. For example, PAR decomposition models require fitting with on-site historical measurements to reduce error margins, as seen in the study by Ma Lu et al. [68].

### 3.2. Photovoltaic modelling platforms

PV modelling platforms are for simulating PV production in APV systems, with a focus on addressing the unique challenges posed by unconventional system designs and effects of crop and microclimate on energy yield. The subsections cover traditional PV modelling tools and their adaptability to APV contexts (3.2.1), simplified approaches for energy yield estimation (3.2.2), and the application of the single-diode model (3.2.3). Additionally, thermal considerations (3.2.4), and customised techniques (3.2.5) are discussed, followed by a review of validation methods (3.2.6). This Section aims to provide insights into accurately estimating PV energy yields in APV settings. Accurate modelling of power production is crucial for the development and evaluation of APV systems, as it enables the conversion of solar irradiance into electrical power. This conversion process is influenced by several factors, such as the efficiency of the PV modules, temperature-related losses, inefficiencies inherent in the system and environmental

**Table 5**

Summary of solar irradiance models, model parameters, validation approaches, and validation results.

| Type of model                                                                      | Modelled parameters                                            | Validation approach                                                             | Validation results                                                                                                                                                                     |
|------------------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shading factor model (Matlab®) [44]                                                | Ground shading conditions                                      | Validation against outputs from PVsyst® and SketchUp® Field                     | RMSE values lower than 0.01 across three APV configurations<br>Error deviations of $\pm 8\%$ between simulated and monthly measured shading levels                                     |
| 3D irradiance model [72]                                                           | Monthly average shading levels of PAR                          | measurements at an overhead APV site for a pear orchard in Belgium              | Diffuse solar fraction of $0.46 \pm 0.01$ (actual value 0.43, relative difference of 6.5 %)<br>Global irradiance overestimated by +7.23 %, diffuse component underestimated by -8.32 % |
| Diffuse solar fraction model [74]                                                  | Diffuse solar fraction under tilt-mounted APV systems          | Validation against Dupraz et al. [2] and a ray-tracing model                    | Relative errors lower than 10 % for all cases                                                                                                                                          |
| Combined shading and irradiance simulation tool [75]                               | Global and diffuse irradiance                                  | Field measurements                                                              | nRMSE between 27 % and 43 %; best model performance (YANG2) $R^2$ of 0.85, nRMSE between 23.8 % and 25.9 %                                                                             |
| 3D model (Solidworks®, Autodesk®) [73]                                             | Shading ratios at different points and seasons<br>Diffuse PAR  | Actual values comparison in an open-field APV system in China                   | Model slightly overestimated PPFD, peak frequency between -6 and -3                                                                                                                    |
| Global irradiance decomposition models [68]                                        |                                                                | Comparison against actual PAR measurements in three Swedish locations           | Generally underestimated solar irradiance, up to $150 \text{ W m}^{-2}$ discrepancy<br>Observed average PAR ratio of 0.47 vs. simulated 0.51                                           |
| Solar spectrum model [76]                                                          | PPFD under PV modules                                          | Actual PPFD measurements under various weather conditions and PV configurations | Pearson correlation coefficient higher than 0.89 for all PV configurations                                                                                                             |
| CFD model [77]                                                                     | Solar irradiance on PV modules                                 | Sensor measurements in APV systems                                              | The average daily accumulated PPFD values derived from the simulations are 26 % larger than the measurements.                                                                          |
| Spatial and temporal PAR variation model [33]                                      | Spatial and temporal variation of PAR                          | Validation against experimental data from Al-Agele et al. [78]                  | $R^2$ error of 0.88, maximum relative RMSE of 17 %                                                                                                                                     |
| Spatio-temporal PPFD model [79]                                                    | Spatially predicted PPFD                                       | k-fold cross-validation with observed PPFD values                               | General excellent agreement, slight overestimation during the last hours of sunlight                                                                                                   |
| Spatio-temporal PPFD model [80]                                                    | PPFD under a semi-transparent PV module                        | Validation against multiple PPFD sensor measurements.                           | Relative error of 1 % on cumulative daily irradiation under contrasting sky conditions.                                                                                                |
| Algorithm for estimating cumulative irradiance radiation [81]                      | Cumulative global irradiance in greenhouses                    | Measurements within a greenhouse with 50 % PV coverage                          |                                                                                                                                                                                        |
| Shadowing model for solar irradiance [82]                                          | Solar irradiance in different locations inside a PV greenhouse | Actual measurements comparison                                                  |                                                                                                                                                                                        |
| PASE 1.0 light model (Python-based with ray casting and geometric projection) [83] | Irradiance on PV modules and crop canopy                       | Validation against field data from a ground-mounted APV system in Belgium       |                                                                                                                                                                                        |



conditions, as well as the spectral response, low-light response of the PV modules, and reflection losses (angle of incidence).

APV systems differ from conventional PV systems in several ways that can affect power production. They are designed to facilitate crop growth, often employing unconventional designs such as elevated panels, vertically mounted rows with large inter-row distances, or specialised modules with increased cell gaps or spectrally selective properties. An example of such a design is presented in Fig. 10.

These designs improve growing conditions but may simultaneously affect the environmental conditions of the PV modules, such as POA irradiance, ambient temperature, or wind speed, leading to different power outputs. Additionally, interactions between crops and panels influence energy output through variations in ground albedo and soiling, adding layers of complexity to APV system performance modelling. Ground albedo, or the reflectivity of the surface beneath the panels, can significantly influence the performance of bifacial PV modules by affecting rear-side irradiance. High-albedo surfaces, such as snow (albedo  $\sim 0.67$ ), enhance energy conversion compared to lower-albedo surfaces like crops (e.g., winter wheat at 0.25 or ley grass at 0.19) [84]. Seasonal and crop-specific variations further amplify the need for dynamic albedo modelling to capture these changes accurately. Johansson et al. [84] demonstrated that dynamic models can improve the accuracy of albedo predictions. Using engineered ground materials with higher albedo offers another strategy to enhance PV performance. The importance of incorporating dynamic albedo modelling into PV performance simulations was further highlighted by Ma Lu et al. [85]. Their validation of the AgriOptiCE® framework demonstrated that dynamic albedo values provide greater accuracy compared to fixed or satellite-derived albedo inputs, achieving  $R^2 > 0.85$  for irradiance predictions in APV systems located in Sweden and the United States. Fixed albedo values often led to errors, especially during snow events or periods of significant crop reflectivity changes. These findings align with broader research by Ma Lu [86], which emphasises the dynamic nature of albedo in APV systems. Seasonal changes, crop reflectivity, and shading effects necessitate crop-specific albedo models to better predict energy yields and optimise system configurations.

In addition to albedo, soiling presents another variable that impacts energy yield in APV systems. Dust and organic material deposition on PV modules can reduce efficiency, and this effect is influenced by agricultural practices, irrigation, and increased human activity in APV setups. While less studied than albedo, soiling is expected to vary significantly

between APV and conventional PV systems due to these unique conditions [6,72,87]. The diversity of APV designs, ranging from vertical to tilted structures, further complicates the impact of soiling, making it an important area for future research.

### 3.2.1. Traditional photovoltaic modelling tools and their use in agrivoltaic systems

Many established software tools and open-source models are available for modelling conventional PV systems. Among them, PVWatts, PVsyst, pvlb and the System Advisor Model (SAM) are commonly used [36]. In the absence of APV-specific energy models, these conventional tools have been deployed. PVWatts [88], for example, was used by Jing et al. [49] to estimate the energy yields of rooftop APV systems in Shenzhen, China, accounting for factors such as module type, system design, and climate data. Willockx et al. [72] also employed PVWatts to simulate the power output of a 13.3 kW<sub>p</sub> PV system elevated above pear orchards in Belgium. They validated the results against measured power output and obtained the following statistical metric values: Mean Absolute Error (MAE) of 50 W, RMSE of 100 W, and mean relative error per kW<sub>p</sub> of 3.6 W.

Malu et al. [90] used the flexibility of SAM's open-source code for energy modelling for their specific system requirements. Their simulations focused on a one-acre APV grape farm in Nagpur, India. They optimised the tilt angle and predicted monthly energy conversion using SAM, which they then extrapolated to estimate a generation of over 16 TWh annually if implemented across all grape farms in India. However, their study lacked experimental validation of the projected energy yields. Cavadini and Cook [91] modified SAM to include the thermal effects of green roofs, highlighting the adaptability of this tool. PVsyst® has been used to model the energy yields of an elevated APV system integrated with rice cultivation, where comparisons between measured and modelled data showed an overestimation of 4.87 % [92]. It has also been applied to model the electricity generation of different APV bifacial configurations for various locations in the United Kingdom [93,94]. Expanding on SAM's potential, Barron-Gafford et al. [95] demonstrated its utility in APV by quantifying the synergy between co-located agricultural and PV systems in dryland regions. Using SAM alongside empirical data, they highlighted improvements in PV efficiency through reduced panel temperatures attributed to plant transpiration. This cooling effect enhanced PV generation by approximately 3 % during peak growing months and 1 % annually. Riaz et al. [96] used the Sandia



Fig. 10. Presents the last step in the building process of an elevated APV system for a pear orchard with increased PV cell spacing to improve light availability [72].



array performance model from pvlib Python's Library [97] to model the energy output of two APV system configurations: a monofacial south-facing tilted array and a vertical bifacial east-west facing array. They compared these layouts with two different density configurations under varying irradiance conditions for two days. The measured data were reported in the study by Valle et al. [98], and the simulations slightly overestimated the measurements.

Traditional PV modelling tools simplify electrical yield modelling using frameworks that have been validated for conventional PV systems, allowing for yield analysis without the need for custom programming. However, as these tools were originally developed with a focus on conventional PV systems, they lack features such as combined energy and crop yield optimisation, the analysis of trade-offs between the two, microclimate effects on energy conversion, and crop-dependent ground albedos.

### 3.2.2. Direct and simplified approaches for energy yield estimation

Some studies rely on simple energy yield estimation techniques. For instance, Amaducci et al. [22] used a method in which they multiplied the incident irradiance by an assumed static module efficiency of 14 % to predict the energy yield per square metre of PV area in their system. Similarly, Riaz et al. [42] used fixed energy conversion efficiencies of 19 % and 16 % for the front and rear bifacial PV surfaces, respectively. Although these methods are computationally simple, they provide less insight than more detailed electrical models. For example, they do not properly account for losses from partial shading of modules, or differences in PV cell efficiency as a function of cell operating temperature.

The simplest approach used was the use of POA solar irradiance as a proxy for electricity yield without electrical modelling, such as by Riaz et al. [25,96]. They used this approach to investigate the trade-offs between crop and energy yields in optimisation studies of different APV designs. Although still relatively simple approaches, other studies estimated the electricity yield of PV modules by accounting for the incident global irradiance, the ambient or module temperature, and the specific characteristics of the modules, such as efficiency under Standard Test Conditions (STC) and thermal coefficients [58,99]. However, to ensure that such optimisation studies properly reflect APV system outputs, they should be reinforced with electrical modelling to remove designs that

may have high POA irradiance but low power output, such as those with frequent partial shading of PV modules.

### 3.2.3. Single-diode model

The single-diode model provides a more detailed physics-based approach by simulating the central photophysics of the PV cell. It uses an electrical equivalent circuit to represent the current-voltage relationship based on parameters such as photocurrent, saturation current, shunt resistance, and series resistance and is commonly used to model conventional PV systems [100].

Campana et al. [23] optimised a five-parameter single diode model by minimising the error between calculated and measured I-V curves in a dual optimisation study on energy and crop yield in a vertical bifacial PV system in Sweden. Only convective thermal effects were considered, i.e., ambient and solar irradiance temperature dependence and wind speed cooling effect. Fixed albedo was considered, thus ignoring the effects of snow and/or ground cover or vegetative phase of the modelled crops on the PV production. The modelled energy yields were compared to PVsyst® for validation as seen in Fig. 11.

Gao et al. [101] applied single-diode equations for each individual cell in a greenhouse-mounted PV system with tracking functionality in Delft, Netherlands. They modelled bypass diodes and shaded cells to account for the local effects of shading and temperature to study the efficacy of different panel densities and tracking schemes. Rodríguez-Pastor et al. [40] also used single-diode equations when modelling the electricity output of an 11.24 MW<sub>p</sub> single-axis tracking bifacial APV system in Seville, Spain. The system was combined with nine different crop species with different light and irrigation needs to study their effect on energy output. They found an increase of 2.5 kWh kW<sub>p</sub><sup>-1</sup> during months with high ambient temperatures and a decrease in module temperature of up to 1.8 °C due to the presence of crops. They related current and voltage to parameters such as photonic current, which were first derived under STC, before integrating the incoming irradiance with the thermal properties of the module to determine the electrical output. They found that the model presented reasonable accuracy when compared to exported energy, despite not capturing events such as inverter clipping, curtailment or power shutdown due to maintenance tasks. Notably, they stated that a two-diode model would not result in

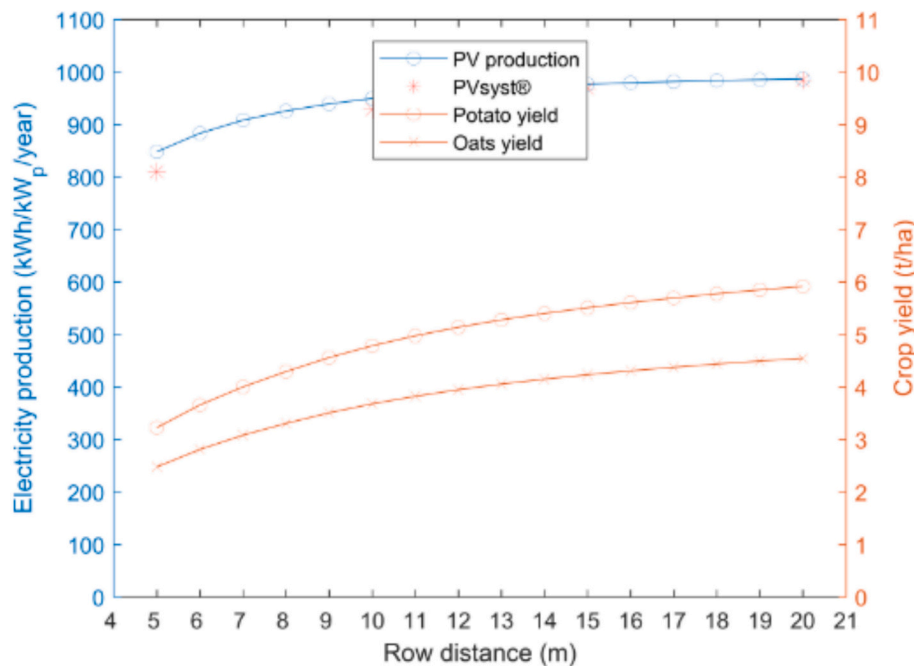


Fig. 11. The relationship between row distance, specific photovoltaic production, and crop yield for both oats and potatoes is examined. Additionally, the modelled specific PV system's production is compared to the commercial software PVsyst® [23].

significant differences. In fact, two-diode models have not been implemented in any of the APV studies analysed. Ghaffarpour et al. [99] implemented a simplified single-diode as part of an integrated model for greenhouse PV systems that accounted for both temperature and irradiance effects on panel performance. Their energy model did not account for factors such as inverter losses and therefore estimated slightly higher power production than PVsyst®.

### 3.2.4. Thermal considerations in photovoltaic efficiency modelling

The effect of cell temperature on PV cell efficiency is important for accurate modelling, as increased PV cell temperature reduces voltage and power output [102]. APV systems may see different cell temperature profiles than conventional PV systems due to factors such as unconventional system designs (e.g., higher height from ground, and thus, higher wind speed) and crop transpiration, further motivating the inclusion of thermal considerations [103].

In some studies, thermal modelling was performed to investigate

changes in module and cell temperature without incorporating it in electrical yield calculations, using Computational Fluid Dynamics (CFD) [77,104], Finite Element Methods (FEM) [105], and analytical methods [106]. Further information can be found in Section 3.3.2.

More often, the influence of temperature is used directly in the modelling of electrical power. For example, Gao et al. [101] used the following empirical relationship [107] to derive module temperature from ambient conditions, and Zhang et al. [108] used it to estimate module temperature:

$$T_m = 0.943 \cdot T_{amb} + 0.028 \cdot I - 1.528 \cdot v + 4.3, \quad (2)$$

where  $T_m$  [°C] represents the module temperature,  $T_{amb}$  [°C] the ambient temperature,  $I$  is the irradiance [ $\text{W m}^{-2}$ ], and  $v$  is the wind speed [ $\text{m s}^{-1}$ ].

Fu and Zhou [109] similarly used another empirical equation in a PV greenhouse study that relates changes in module temperature to weather variables. Other researchers, such as Sojib Ahmed et al. [26]

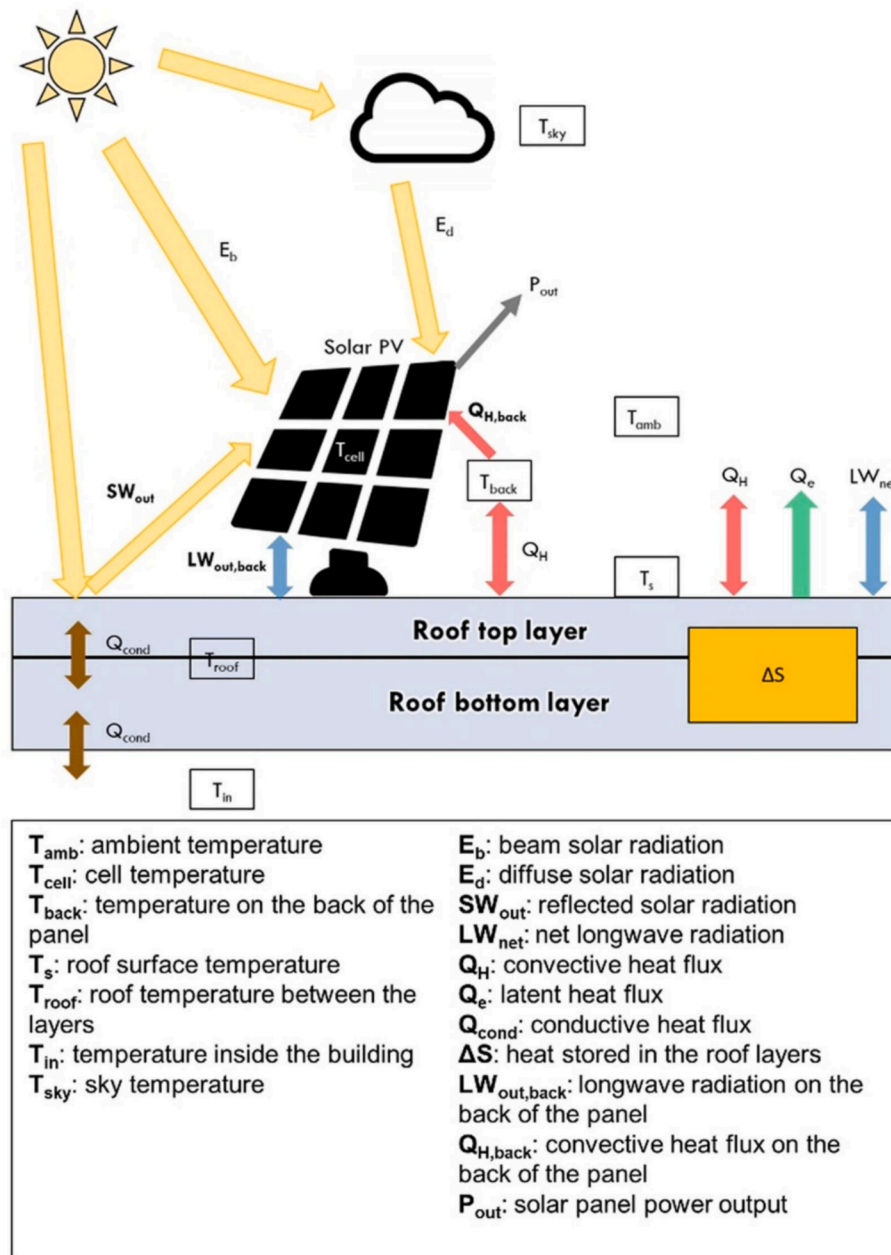


Fig. 12. Heat exchange on a rooftop with PV modules [91].

and Patel et al. [24], used Skoplaki's model [110] to derive cell temperature to improve estimates of panel conversion efficiency. Katsikogiannis et al. [35] used the Sandia National Laboratories (SNL) model [111] to model cell temperatures in several APV designs as a function of irradiance and ambient temperature, while also accounting for system losses such as soiling, module mismatch, and ohmic losses. Ghaffarpour et al. [99] compared two approaches for determining cell temperature in greenhouse-mounted PV systems: an energy balance method and the NOCT (Nominal Operating Cell Temperature) approach. The energy balance method accounted for heat transfer coefficients affected by wind speed, which caused it to predict lower cell temperatures than NOCT calculations. Finally, the King's temperature model [111] has also been used to estimate the PV module temperature in [42,72] for subsequent PV power calculation models.

Some studies consider additional effects beyond cell or panel temperature. Fernández et al. [112] included temperature factors together with Angle of Incidence (AOI), spectral effects, and the active panel area in the efficiency calculation. Cavadini and Cook [91] specifically investigated thermal effects on green roofs through energy balance modelling, depicted in Fig. 12. The model integrates with SAM to evaluate how the surface temperatures of green roofs, made from different materials, affect the conversion efficiency of PV systems. Their model predicted that in Zurich, Switzerland, green roofs could increase annual PV energy yield by 1.8 % on average compared to conventional rooftop systems.

### 3.2.5. Customised techniques

Several studies have developed specialised modelling techniques that focus on tasks that deviate from typical APV energy modelling, such as sub-module energy modelling or Geographic Information System (GIS) based potential studies. These customised methods are employed when the output from traditional approaches, such as single-diode models, does not align with the aim of the research objectives. This is often the case in APV research, where unconventional designs, modules, or the effects of external conditions such as proximity to crops are still being explored.

Sacchelli et al. [113] developed an open-source GIS-model called *r.green.solar* that integrated data sets on terrain, land-use, and spatial constraints to assess the technical energy potential for PV on agricultural land in Italy based on theoretical, legal, technical, practical, and economic factors. The study used PVGIS to assess the theoretical energy yield, but only focused on fixed-ground mounted systems. Trommsdorff et al. [45] used an area-based approach and applied the concept of PV yield per unit area, which is the product of the specific electrical yield ( $\text{kWh kW}_p^{-1}$ ) and the installed capacity per unit area ( $\text{kW}_p \text{m}^{-2}$ ). The specific yield was modelled using Zenit, a tool developed by Fraunhofer ISE, and the density of installed capacity was calculated from the design and spacing of the systems. In a study in the arid Thar Desert of India, the energy conversion of a 105  $\text{kW}_p$  APV system was simply extrapolated to estimate the yields of a 520  $\text{kW}_p$  system. This was done as part of a techno-economic analysis on APV systems, where the government provides subsidies for system sizes of 0.5–2  $\text{MW}_p$  [114]. At the individual cell level, Ravishankar et al. [115] optically modelled Organic Solar Cells (OSC) using light absorption and quantum efficiency to estimate the efficiency parameters and energy yields of semi-transparent solar cells integrated in greenhouses. Perna [116] linked the energy yield to a change in irradiance by adjusting voltage, current and fill factor in a customisable cell model. Finally, Cuppari et al. [117] used a thermodynamic approach that linked weather data such as temperature and wind to module efficiency to model system performance.

### 3.2.6. Validation of photovoltaic models in agrivoltaic settings

Fernández et al. [112] modelled monthly Direct Current (DC) energy yields from PV systems on greenhouses using a research-based code by considering different sources of losses. The outputs were validated against the DC power measurements of an individual opaque mono-

crystalline silicon module installed in Jaén, Spain. They obtained an annual percentual MAE of  $\sim 2\%$ .

Ghaffarpour et al. [99] employed a single equation to model the output DC power of a 3  $\text{kW}_p$  system installed in a PV greenhouse in Iran. This equation establishes a relationship between DC power, incident irradiance, cell temperature, and the electrical and thermal characteristics of the modules. The authors validated the results obtained from this equation by comparing them with those generated by PVsyst® and analysed the system's monthly energy conversion. They observed that the results derived from their equation slightly exceeded those provided by PVsyst®.

Ma Lu et al. [85] validated the effects of using fixed, satellite-derived, and hourly measured albedo on the performance of a vertical bifacial system and a single-axis tracking system using a bifacial photovoltaic model called AgriOptiCE® [23,44,85,118–121]. They compared estimated front and rear irradiances with on-site measurements from a single-axis tracker site in Golden, United States and a vertical APV system as well as traditional PV system power output estimations in Västerås, Sweden, with an  $R^2 > 0.85$  for all the studied cases. A similar study by Willockx et al. [122] evaluated the performance of vertical and single-axis APV systems. Their validation of the vertical system revealed that the simulation model overestimated power production due to shading effects from the mounting structure, which were not accounted for in the model. This underscores the importance of incorporating shading caused by mounting structures into simulation models to enhance the accuracy of energy yield predictions.

### 3.3. Microclimate modelling

The installation of PV panels on agricultural land changes the microclimate, which can affect crop yields and the efficiency of PV panels, as seen in Fig. 5. The microclimate modelling Section examines how APV installations influence local environmental conditions and the implications for both crop productivity and PV system efficiency. As methods for irradiance modelling were discussed in Section 3.1, they are not a focus of this Section. The Section explores methods for assessing thermal conditions (3.3.1), wind patterns (3.3.2), and the combined influence of humidity, rain distribution, and soil moisture (3.3.3). The Section concludes with the validation of microclimate models (3.3.4). By addressing these aspects, this Section demonstrates the significance of integrating microclimatic changes into the design and optimisation of APV systems.

#### 3.3.1. Energy balance and data-driven models for thermal assessment

Energy balance models provide a simplified representation of microclimatic thermal aspects based on transport phenomena and meteorological principles. They are suitable for scenario analysis due to their computational efficiency but are usually dependent on a range of assumptions and simplifications. Statistical regression models based on field measurements link microclimatic parameters with factors such as ground level solar irradiation, which depends on the APV system configuration. However, their accuracy is limited by the amount of data and may not be transferable to other APV systems in other locations.

For closed environments such as greenhouses with integrated PV, energy balance models for heat flow and temperature modelling are often used. Evans et al. [82] developed a 1-D transient heat and moisture transport model for an experimental APV greenhouse setup that accounted for solar irradiance, convection, radiation heat exchange, conduction, and infiltration. The model predicted that the greenhouse air temperatures were generally above ambient throughout the year, demonstrating the potential of such systems to extend the growing season. Ravishankar et al. [115] created a dynamic energy model for greenhouses with semi-transparent organic solar cells. Their simulations demonstrated that net-zero energy greenhouses could be achieved in warm and moderate climates, and that the reduction in sunlight entering the greenhouses was not as significant as the transmittance of the solar



cells would suggest, partly due to the cells replacing the need for shade cloths. Simplified thermal models like the Overall Thermal Transfer Value (OTTV) approach used by Lu et al. [123] have also been used to provide estimates of heat transfer in smaller greenhouse structures. While OTTV offers insights into overall heat transfer, it may oversimplify some aspects of greenhouse microclimates.

Energy balance models have also been used to estimate temperatures in PV systems on green roofs, to analyse long-wave irradiance at the module and ground level, and to analyse large-scale APV systems with bifacial panels [40,91,106]. Shepard et al. [106] developed a model to quantify the downwelling longwave energy at the ground surface in an APV array. Their model demonstrated that longwave and shortwave radiative fluxes can be of similar magnitudes under some conditions, even during daytime. Rodriguez-Pastor et al. [40] created an analytical model for heat transfers through different layers of a bifacial solar panel using electrical circuit elements (Fig. 13). By incorporating parameters like panel geometry, tracking angles, and ground conditions, the model evaluated how changes in system aspects impacted energy yields. Their model, validated with experimental data from a commercial solar power plant, provided regressions for characterising system performance. The regressions achieved  $R^2$  values above 0.85 for both module temperature and energy export in panel strings combined with agricultural crops. However, the model was specifically calibrated for summer months and may require further validation for year-round accuracy. In similar studies, tools such as Matlab [82,91], Engineering Equation Solver (EES) and python [40] are frequently used to solve the equations in energy balance models.

Statistical regression models derived from measured observations have been used to relate climate metrics such as temperature, wind, and rear-surface temperatures of PV modules to plant heat stress [124].

While analytical energy balance models and data-driven methods can be computationally efficient and provide overarching system insights, they normally do not capture the system's response to change, especially for three-dimensional and temporal changes. Such details can be better explored with CFD models.

### 3.3.2. Computational fluid dynamics modelling for microclimate analysis

CFD models excel in the representation of 3D temperature and wind dynamics by numerically solving fluid physics and heat transfer equations to enable high-resolution simulations of localised effects. Numerous studies have used CFD tools to evaluate the temperatures and wind patterns in different APV designs. Based on the findings of this review, every study that has performed CFD analysis on wind conditions in APV systems has also in some way examined the effects on temperature.

Tools such as Solidworks Flow Simulation and ENVI-met were used for integrated, CAD-supported CFD analyses. Johansson et al. [104] and Zainali et al. [77] used Solidworks Flow Simulation to evaluate temperature and wind parameters in a vertical APV system in Sweden. Both validated their models with data from a pilot system in Sweden, performing a transient analysis at intervals of five and ten minutes, respectively. Zainali et al. [77] observed a slight decrease in wind speed

from  $5 \text{ m s}^{-1}$  to  $4 \text{ m s}^{-1}$  at 0–0.5 m heights within the vertical APV system and a modest air temperature rise of  $0.3 \text{ }^{\circ}\text{C}$  due to heat dissipation from PV modules (Fig. 14). Johansson et al. [104] found that vertical bifacial PV modules maintained lower average module temperatures than conventional tilted modules. Both studies were limited by relatively short validation periods of about one day.

Minanda et al. [125] also used Solidworks Flow Simulation for CFD simulations, but for a PV greenhouse system. They modelled air as a steady, turbulent, and incompressible 3D fluid, considering various ventilation combinations. Their results demonstrated that a combination of ventilation, horizontal air flow fans, and exhaust fans could effectively maintain optimal temperature and humidity in a greenhouse with 51 % PV shade coverage. The study provided detailed design parameters but lacked real-world experimental validation. Teng et al. [126] utilised ENVI-met to simulate a rooftop APV system, complementing their simulations with a physical prototype. Their findings revealed that PV panels with crops underneath experienced lower operating temperatures, leading to improved efficiency. On sunny days, PV temperatures were on average  $2.83 \text{ }^{\circ}\text{C}$  lower with crops, resulting in an estimated 1.13–1.42 % efficiency improvement. The physical prototype corroborated these results, showing 3.05 % and 3.2 % more power generation with crops on sunny and cloudy days, respectively. However, the study was limited by its short timeframe and imperfect weather-data matching between simulations and experiments.

As for the tools specialised in CFD, Ben Amara et al. [127] used ANSYS Fluent to model the wind flow inside a greenhouse with and without PV panels on the roof, accounting for turbulence using the  $k-\epsilon$  model and natural convection effects. The simulations were performed for temperature and humidity for individual days representing summer and winter conditions. The model was validated against experimental data, showing good agreement with absolute errors for temperature ranging from  $0.3 \text{ }^{\circ}\text{C}$  to  $2.19 \text{ }^{\circ}\text{C}$  and for relative humidity from 0.61 % to 8.42 % across seasons. Their results showed that PV panels decreased solar irradiation by 65 % and reduced internal air temperature by  $3\text{--}5 \text{ }^{\circ}\text{C}$  on clear days while increasing relative humidity by  $\sim 12 \text{ }^{\circ}\text{C}$ . Williams et al. [128] developed a more comprehensive ANSYS Fluent model for an open-field APV system, investigating the effects of panel height, ground albedo, and evapotranspiration on solar module temperatures and the surrounding microclimate. Their model, validated against experimental data from a solar farm in Sarnia, Ontario, demonstrated high accuracy with an RMSE of  $1.91 \text{ }^{\circ}\text{C}$  for module surface temperatures. They found that increasing panel heights from 0.5 m to 4.5 m, increasing ground albedo from 0 to 1, and including evapotranspiration all contributed to solar farm cooling. Notably, they observed that an APV system mounted 4 m above soybeans could achieve up to  $10 \text{ }^{\circ}\text{C}$  reduction in panel surface temperature compared to a traditional solar farm mounted 0.5 m above bare soil. However, the study has some limitations: simulations were conducted for a single day in July, thereby missing any potential seasonal variations; it assumed a constant evapotranspiration rate, which may vary with crop type, growth stage, and environmental conditions; and it did not consider different crop species, which could have varying impacts on the microclimate. Rane

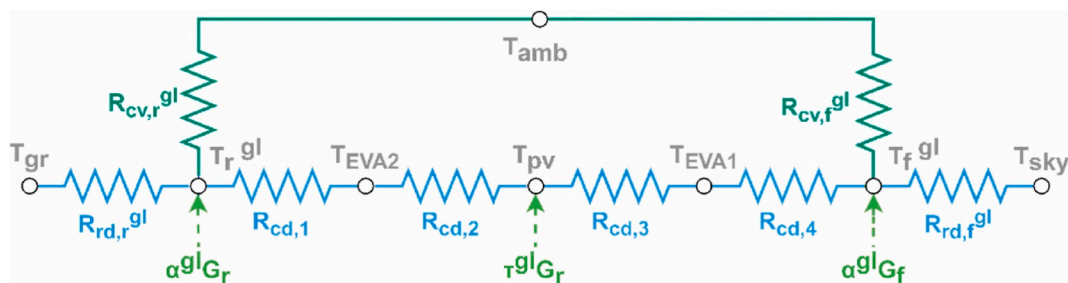


Fig. 13. The electrical analogy for the thermal model of bifacial panels [40]. Resistors represent thermal pathways (conduction, convection, and radiation), T points show temperatures at different layers, and  $\alpha^{\text{gl}}_{\text{Gr}}$  and  $\tau^{\text{gl}}_{\text{Gr}}$  terms represent absorbed and transmitted solar irradiance.

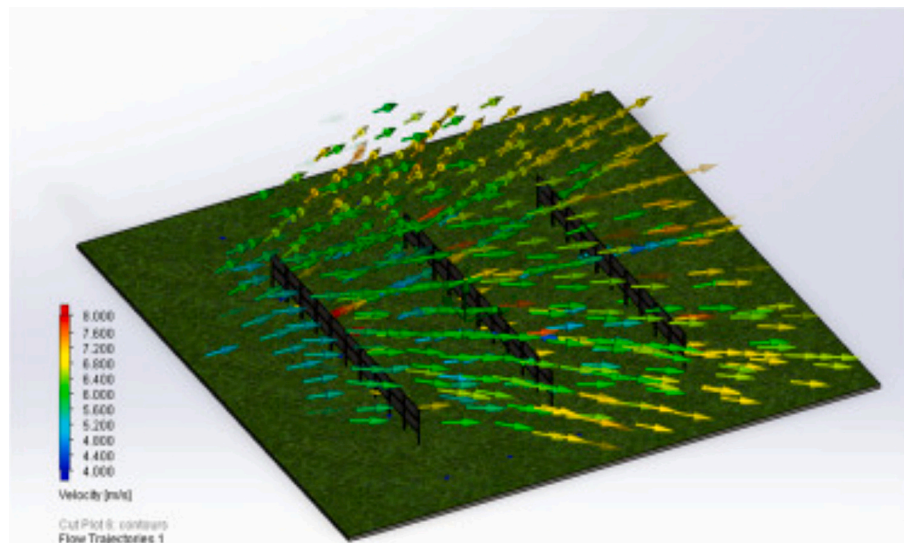


Fig. 14. CFD model computation of wind speed and direction [77].

et al. [105] developed a FEM using COMSOL Multiphysics to simulate the airflow and thermal environment within an elevated APV system with shading nets. The model combines airflow in porous plant domains and free air domains, including surface-to-surface radiation and moisture transport. Simulations showed minimal temperature differences in plants with and without PV modules (25.06–28.2 °C vs 25.07–28.5 °C). For validation, they used thermal infrared imaging of crepe jasmine leaves, finding temperatures of 31–34 °C at 33 °C ambient. While this provides some basis for comparing shading effects, the model's reliance on literature-derived parameters for plant properties and limited validation with a different plant species than simulated are potential limitations of this approach.

### 3.3.3. Air humidity, soil moisture, and rain distribution modelling

In APV systems, moisture and humidity are closely linked to parameters such as wind, temperature, solar irradiance, and plant growth. As these are interdependent parameters, they are rarely modelled in isolation. Additionally, the modelling of precipitation distribution on the ground, which is closely linked to soil moisture, is still a relatively understudied aspect in the APV field. The presence of PV modules in cultivated areas can lead to considerable spatial heterogeneity in rainfall distribution on the ground. Accurate modelling of this phenomenon can help optimise irrigation practices and develop effective strategies for tracking-based PV installations during rainfall events to improve the uniformity of rainfall distribution. This section addresses the different modelling approaches used to study these parameters, from mechanistic methods to computational techniques.

In research primarily concerned with water dynamics in APV systems, mechanistic models are often used. Such models aim to mathematically represent the processes associated with energy, water balance, and plant physiology. For example, Marrou et al. [129] used an energy balance model that accounts for transpiration and evaporation based on climatic requirements, land cover, and surface conductivity determined by field measurements. Elamri et al. [130] presented AVirrig, a holistic model for assessing plant growth and water balance that emphasises the response of stomatal conductance to light and hydration. Wu et al. [131] adopted a composite modelling tactic in which they combined sub-models for irradiance, rain interception, evapotranspiration, and soil hydration, allowing the simulation of impacts over a century. While Marrou et al. [129] and Elamri et al. [130] investigated short-term fluctuations, Wu et al. [131] focused on persistent patterns. These studies show the importance of combined modelling of energy, water, and plant physiology over different time periods for an improved

understanding of air and soil moisture in APV systems.

Studies aiming at a simpler integration of changes in humidity and soil moisture content often use empirical methods or models that combine empirical and theoretical approaches (such as the FAO Penman–Monteith equation) to assess changes in water availability for plant growth. Marrou et al. [129] also calculated reference evapotranspiration using Penman–Monteith to determine irrigation requirements for lettuce as part of their more integrated approach. Amaducci et al. [22] used Penman–Monteith together with a model for photosynthesis and transpiration (GECROS v3.0) to model leaf transpiration and temperature. The coupled modelling approaches that integrate Penman–Monteith and other similar models show how evapotranspiration can be linked to other climatic inputs to more accurately represent soil moisture conditions. Campana et al. [132] estimated the evapotranspiration of potatoes and oats using the Hargreaves and Samani [133] relationship but using mesoscale data in a Swedish case study. The evapotranspiration model was coupled with a model depicting ground-surface irradiation to account for the effect of shading on evapotranspiration. No other micro-climatic effects were considered in the computation of the reference evapotranspiration, such as wind speed and direction variations, or ground temperature variations.

On the computational front, CFD techniques have shown promise in solving 3D moisture transport. Ben Amara et al. [127] used ANSYS Fluent to model the distribution of water vapor in a greenhouse with roof-integrated PV panels. By incorporating evapotranspiration via user-defined functions, they captured humidity dynamics across different seasons. This level of physical accuracy is valuable but can involve complicated settings and high computational requirements.

The effects of PV modules on rainfall distribution on the ground were investigated by Elamri et al. [134]. They presented AVrain, a 2D model that characterises the interception and redistribution of rain by PV modules that was validated with three different rain events and with different PV module configurations (Fig. 15).

It identified sheltered areas and zones where the actual rainfall exceeds that of a reference cropland (without PV modules). The model requires as inputs the geometrical dimensions of the PV structure, such as the height of the modules, the orientation and the tilt angle as well as the distance between the supporting columns of the PV frame system. To simulate the rain distribution, the model uses meteorological data (rain intensity, wind speed and wind direction) with a resolution of one minute. The output of AVrain, i.e., the spatial distribution of rainfall, can be used as input in software packages such as HYDRUS-2D [79] to simulate soil moisture.

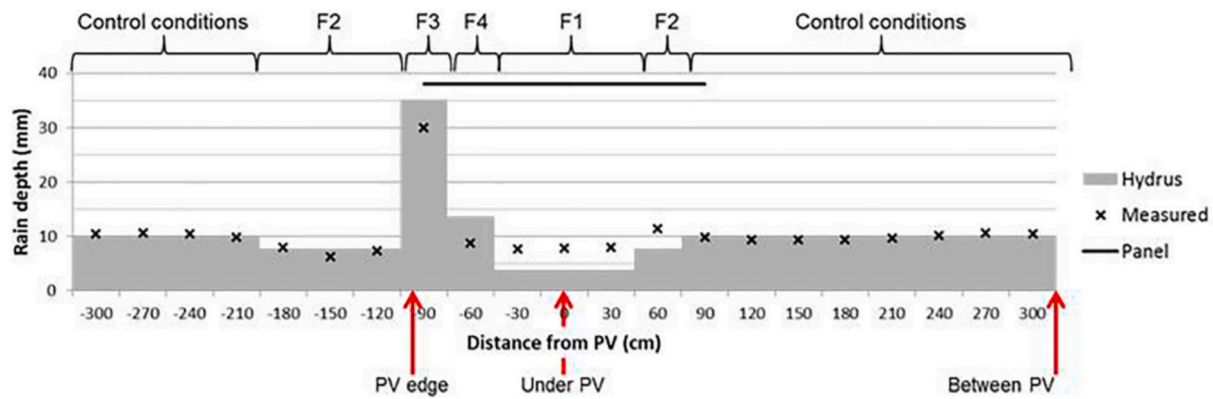


Fig. 15. Time-dependent upper boundary conditions were applied in Hydrus-2D for the analysed rain event, during which the panel tilt angle was adjusted to reduce rain interception as part of an avoidance strategy [134].

### 3.3.4. Validation of microclimate models

In APV systems, microclimate models are crucial for understanding how localised conditions affect both plant growth and solar panel efficiency. Table 6 provides a detailed summary of various studies focused on the validation of microclimate models in APV systems. These studies compare simulated data to actual measurements under different environmental conditions and model configurations.

For instance, the synthetic model used for soil temperature and moisture validation in Gansu, China, showed a RMSE of 2.45 °C for soil temperature, with an underestimation observed in summer. Meanwhile, the Python-based stochastic model demonstrated a strong correlation ( $R^2 = 0.98\text{--}0.99$ ) for air temperature and wind speed in Oregon and North Carolina, indicating its robustness in simulating daily averages. However, challenges remain in simulating extreme conditions, as seen in the CFD model for air temperature and relative humidity in a PV greenhouse, where errors ranged from 0.98 to 2.19 % for air temperature in winter and 0.38–1.37 % in summer. Additionally, while the thermal and moisture transport model showed relatively small errors in air temperature (1.9 °C max and 1.5 °C min), moisture predictions still exhibited some over/underestimations. These examples highlight that while the models perform well in typical conditions, improvements are needed to handle extreme environmental factors more accurately. Future advancements may focus on refining model sensitivity to such extremes and improving their integration of complex data inputs to enhance predictive capabilities across diverse APV settings.

### 3.4. Crop modelling in agrivoltaic systems

The crop modelling Section evaluates the tools and methodologies used to predict the impact of APV systems on agricultural productivity. Subsections include discussions on empirical modelling methods (3.4.1), followed by considerations for validations (3.4.2). The analysis focuses on how different modelling strategies simulate the effect of shading, microclimatic changes, and light distribution.

In the field of crop modelling, mechanistic and empirically derived models serve as critical tools to simulate the growth and development of plants over time [149]. These models incorporate a wide range of factors, including environmental conditions (temperature, irradiance, and humidity), soil properties, and plant-specific traits, to predict plant responses to varying conditions [140]. Through mathematical equations or algorithms, they capture essential physiological processes such as photosynthesis, respiration, nutrient uptake, and biomass accumulation. By integrating these processes with plant developmental stages, from germination to maturity, crop models provide insights into growth dynamics, aid in optimising agricultural practices, and support decision-making under diverse scenarios [141].

Despite their utility, crop growth models are inherently

Table 6

Summary of the studies addressing microclimate validation in APV systems.

| Type of model                             | Modelled parameters                  | Validation approach                                                                                | Validation results                                                                                                                                               |
|-------------------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Synthetic model [131]                     | Soil temperature, soil moisture      | Validation against measurements in four zones of an APV system, Gansu, China, April–September 2016 | Soil temperature: RMSE = 2.45 °C, underestimation in summer; soil moisture: errors negligible with an RMSE = 0.02 $\nu/\nu$ (volume of water per volume of soil) |
| Python-based stochastic model [117]       | Air temperature, wind speed          | Historical weather data validation in Oregon and North Carolina, USA                               | Strong correlation ( $R^2 = 0.98\text{--}0.99$ ) for daily averages; slightly lower for extremes                                                                 |
| CFD model [127]                           | Air temperature, relative humidity   | Measurements in a PV greenhouse, INRAE, France, winter and summer                                  | Winter: absolute relative humidity error 2.75–8.42 %, air temperature 0.98–2.19 %; summer: relative humidity error 0.61–1.71 %, air temperature 0.38–1.37 %      |
| CFD model [104]                           | Module temperature                   | Validation against thermal camera and other models, Kärrobo Prästgård, Sweden, September 25, 2021  | Generally, underestimates module temperature by 3–4 °C; good agreement with other models                                                                         |
| Thermal and moisture transport model [82] | Air temperature, relative humidity   | Measurements in a simulated PV greenhouse, Pennsylvania, USA, 2020–21                              | Air temperature: errors of 1.9 °C and 1.5 °C for max and min temperatures; moisture: over/underestimations, improved with air temperature data incorporation     |
| CFD model [77]                            | Soil temperature, module temperature | Comparisons with measurements, Kärrobo Prästgård, Sweden, June 23, 2022                            | Soil temperature errors <1 °C; module temperature closely matches other model predictions                                                                        |
| Analytical model [134]                    | Spatial distribution of rain         | Validation with natural and simulated rainfall measurements, Montpellier, France, 2015–16          | Linear correlation slope between 0.88 and 1.45; RMSE 0.03 to 0.22, some overestimations                                                                          |



simplification of complex biological systems. Their accuracy depends heavily on the quality of input data and assumptions made during model development. Environmental factors, which directly influence crop growth, are particularly critical in determining model reliability. Key variables include climate factors such as temperature, precipitation, solar irradiance, and CO<sub>2</sub> levels, which govern photosynthesis and other physiological processes. Soil characteristics, such as nutrients availability, water retention capacity, and texture, while management practices like irrigation, fertilization, and pest control modify growing conditions and productivity outcomes [141]. The unique challenges posed by APV systems underscore the need for adaptations to conventional crop growth models. APV systems alter the microclimate by introducing spatial and temporal heterogeneity in light distribution, temperature, humidity [147]. Light reduction beneath solar panels, in particular, has significant effects on plant growth, with ongoing research aimed at quantifying these impacts. The interaction between crops and solar panels creates complex dynamics that current models may not fully capture. Additionally, APV-specific management practices, such as optimising panel configurations, tailoring irrigation strategies, and accounting for shading effects further complicate accurate simulation. Enhancing crop growth models to reflect these unique characteristics is essential for improving their predictive capabilities in APV systems.

A significant challenge in this context is the limited availability of detailed field data from APV systems, which hinders model calibration and validation. Addressing this data bottleneck requires collaboration among researchers, model developers, and practitioners to generate robust datasets and refine modelling approaches. Mechanistic models, which use differential equations to represent underlying processes, are well-suited for simulating complex interactions and climate change impact but demand extensive input data. Conversely, empirical models, while less data-intensive, are limited to specific conditions and lack the generalisability of mechanistic approaches. Crop growth models have long been utilised for yield forecasting, growth analysis, and decision support by integrating data from disciplines such as crop phenology, soil water hydrology, and climate science [135]. Their ability to offer cost-effective alternatives to field experiments makes them invaluable for exploring management practices and environmental impacts [136]. However, publicly available models currently lack features designed specifically for APV systems. Addressing this gap will require adapting existing platforms or developing new ones capable of capturing the unique microclimate and management interactions inherent to APV systems.

A range of popular and adaptable crop modelling platforms exists, each varying in their mechanistic or empirical foundations and their ability to simulate specific crops. While none are explicitly tailored for APV systems, these tools provide a foundation for researchers to explore the integration of APV specific dynamics into crop modelling, paving the way for more accurate and actionable insights in this emerging field. These models integrate knowledge from diverse disciplines such as crop phenology, soil water hydrology, and nutrient cycling, providing a robust framework for yield forecasting and decision support systems. Table 7 summarises the key outcomes, challenges, and way forward for several widely used crop simulation models.

Each model addresses specific needs, using different methodologies and data inputs to provide accurate predictions. Despite their strengths, these models face common challenges that need to be addressed to enhance their effectiveness. Models like DSSAT, APSIM, CropSyst, and AquaCrop provide comprehensive simulations that integrate weather, soil, and management practices, while SPASS, APSIM, and STICS are highly modular, allowing adaptation for various crops and conditions [138,140,141]. Specialised applications are evident in models such as AquaCrop, which excel in water stress simulations and sustainable land use, respectively. However, the complexity and extensive data requirements of models like DSSAT, APSIM, TOMGRO and CERES pose barriers, particularly in data-scarce contexts. Furthermore, models like STICS and AquaCrop exclude certain critical processes such as nutrient

**Table 7**

Key outcomes, challenges, and future directions for crop models.

| Model          | Key Outcomes                                                                                                                            | Challenges                                                                                                                            | Future Directions                                                                                                                                                                                                                                                                                                                                                                      |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| APSIM [137]    | <ul style="list-style-type: none"> <li>• Customisable</li> <li>• Extensive application</li> <li>• Robust performance</li> </ul>         | <ul style="list-style-type: none"> <li>• Data-intensive</li> <li>• Complex evaluations</li> <li>• Limited capabilities</li> </ul>     | <ul style="list-style-type: none"> <li>• Expand capabilities</li> <li>• Improve accessibility</li> <li>• Facilitate user collaboration</li> <li>• Integrate soil salinity</li> <li>• Improve nutrient stress modelling</li> </ul>                                                                                                                                                      |
| AquaCrop [138] | <ul style="list-style-type: none"> <li>• Robust water stress simulations</li> <li>• Intuitive interface</li> </ul>                      | <ul style="list-style-type: none"> <li>• Lack nutrient cycle simulations</li> </ul>                                                   | <ul style="list-style-type: none"> <li>• Enhance stress simulations</li> <li>• Improve user-friendliness</li> </ul>                                                                                                                                                                                                                                                                    |
| CERES [139]    | <ul style="list-style-type: none"> <li>• Emphasises soil–water balance</li> <li>• Integrates grazing effects</li> </ul>                 | <ul style="list-style-type: none"> <li>• Extensive data needs</li> <li>• May miss some environmental interactions</li> </ul>          | <ul style="list-style-type: none"> <li>• Add detailed sub-models</li> <li>• Improve interactions</li> <li>• Validate long-term</li> <li>• Include more factors</li> <li>• Simplify data</li> <li>• Continuous validation</li> </ul>                                                                                                                                                    |
| CropSyst [140] | <ul style="list-style-type: none"> <li>• Versatile</li> <li>• User-friendly</li> <li>• Robust simulations</li> </ul>                    | <ul style="list-style-type: none"> <li>• High data requirements</li> <li>• Simplified processes</li> </ul>                            | <ul style="list-style-type: none"> <li>• Improve responsiveness</li> <li>• Refine algorithms</li> <li>• Diverse validation</li> <li>• Simplify use</li> <li>• Expand calibration datasets</li> </ul>                                                                                                                                                                                   |
| DSSAT [141]    | <ul style="list-style-type: none"> <li>• Comprehensive integration</li> <li>• Robust long-term performance</li> </ul>                   | <ul style="list-style-type: none"> <li>• High complexity</li> <li>• Extensive data needs</li> <li>• Limited responsiveness</li> </ul> | <ul style="list-style-type: none"> <li>• Improve responsiveness</li> <li>• Refine algorithms</li> <li>• Diverse validation</li> <li>• Simplify use</li> <li>• Expand calibration datasets</li> </ul>                                                                                                                                                                                   |
| EPIC [142]     | <ul style="list-style-type: none"> <li>• Comprehensive</li> <li>• Reliable long-term predictions</li> </ul>                             | <ul style="list-style-type: none"> <li>• Overestimates nitrogen impact</li> <li>• Limited real-time use</li> </ul>                    | <ul style="list-style-type: none"> <li>• Improve responsiveness</li> <li>• Refine algorithms</li> <li>• Diverse validation</li> <li>• Simplify use</li> <li>• Expand calibration datasets</li> </ul>                                                                                                                                                                                   |
| GECROS [143]   | <ul style="list-style-type: none"> <li>• Mechanistic model</li> <li>• High temporal resolution</li> <li>• Stable grain yield</li> </ul> | <ul style="list-style-type: none"> <li>• Data-intensive</li> <li>• Complex for novices</li> </ul>                                     | <ul style="list-style-type: none"> <li>• Improve responsiveness</li> <li>• Refine algorithms</li> <li>• Diverse validation</li> <li>• Simplify use</li> <li>• Expand calibration datasets</li> </ul>                                                                                                                                                                                   |
| SALUS [144]    | <ul style="list-style-type: none"> <li>• Integrates sustainability</li> <li>• Suitable for diverse regions</li> </ul>                   | <ul style="list-style-type: none"> <li>• Complex</li> <li>• Needs extensive calibration data</li> </ul>                               | <ul style="list-style-type: none"> <li>• Simplify calibration</li> <li>• Expand database</li> <li>• Improve usability</li> <li>• Add nutrient and stress simulations</li> <li>• Enhance phenology</li> </ul>                                                                                                                                                                           |
| SIMPLE [145]   | <ul style="list-style-type: none"> <li>• Simple</li> <li>• Flexible</li> <li>• Well-suited for educational purposes</li> </ul>          | <ul style="list-style-type: none"> <li>• Lacks detailed dynamics</li> <li>• Fixed assumptions</li> </ul>                              | <ul style="list-style-type: none"> <li>• Add nutrient and stress simulations</li> <li>• Enhance phenology</li> </ul>                                                                                                                                                                                                                                                                   |
| SPASS [146]    | <ul style="list-style-type: none"> <li>• Highly modular and flexible</li> <li>• User-friendly interface</li> </ul>                      | <ul style="list-style-type: none"> <li>• Complex validation</li> <li>• Windows-specific</li> <li>• Data-intensive</li> </ul>          | <ul style="list-style-type: none"> <li>• Expand operating system support</li> <li>• Enhance validation</li> <li>• Refine parameters</li> <li>• Integrate detailed factors</li> <li>• Improve accuracy</li> <li>• Validate broadly</li> <li>• Validation with different tomato cultivars</li> <li>• Code implementation in widely used programming languages, such as Python</li> </ul> |
| STICS [147]    | <ul style="list-style-type: none"> <li>• Modular</li> <li>• Adaptable</li> <li>• Effective simulations</li> </ul>                       | <ul style="list-style-type: none"> <li>• Excludes some processes</li> <li>• Simplified growth stages</li> </ul>                       | <ul style="list-style-type: none"> <li>• Improve accuracy</li> <li>• Validate broadly</li> <li>• Validation with different tomato cultivars</li> <li>• Code implementation in widely used programming languages, such as Python</li> </ul>                                                                                                                                             |
| TOMGRO [148]   | <ul style="list-style-type: none"> <li>• High physiology detail</li> <li>• Flexible</li> <li>• Adaptable</li> </ul>                     | <ul style="list-style-type: none"> <li>• Specific for tomatoes</li> <li>• Needs calibration</li> </ul>                                | <ul style="list-style-type: none"> <li>• Validation with different tomato cultivars</li> <li>• Code implementation in widely used programming languages, such as Python</li> </ul>                                                                                                                                                                                                     |
| WOFOST [149]   | <ul style="list-style-type: none"> <li>• Robust ecophysiological simulations</li> <li>• Adaptable parameters</li> </ul>                 | <ul style="list-style-type: none"> <li>• Inconsistent implementations</li> <li>• Limited nutrient dynamics</li> </ul>                 | <ul style="list-style-type: none"> <li>• Ensure consistency</li> <li>• Expand stress response simulations</li> </ul>                                                                                                                                                                                                                                                                   |

cycling and pest impacts, affecting their comprehensive applicability [138,147]. To move forward, enhancing usability and accessibility by simplifying data requirements, improving user interfaces, and expanding operating system support is essential. Integrating additional simulations routines for biotic and abiotic factors, such as nutrient dynamics and pest impacts, will enhance accuracy and robustness. Continuous

validation and calibration using diverse datasets and involving end-users for practical refinements are crucial. Promoting consistency across implementations and expanding stress response simulations will further improve their reliability. By addressing these challenges and implementing suggested improvements, crop simulation models can be further enhanced to support sustainable agricultural practices and effectively assess the impacts of climate change on crop production, ensuring they remain indispensable resources for researchers, policy-makers, and farmers, thereby contributing to global food security and environmental sustainability.

#### 3.4.1. Research in crop modelling for agrivoltaic systems

Several studies have utilised crop models to analyse the effects of APV systems on crop growth. Dupraz et al. [2] pioneered the use of crop models in APV studies by integrating an irradiance model with the STICS crop model [147] to assess the impact of shading on wheat, finding yield reductions of 19 % under full-density and 8 % under half-density scenarios compared to open-field conditions. Cossu et al. [150] adopted a simplified approach, assuming a 1 % increase in additional light leads to a 1 % increase in yield. They classified crops based on the Daily Light Integral (DLI) into three categories. By assessing 14 horticultural and floricultural crops in commercial PV greenhouses with coverage ratios ranging from 25 % to 100 %, they found that all crops could be grown under 25 % PV coverage. The study also identified varying light thresholds for medium and low light-demanding crops. Varo-Martínez et al. [56] also implemented a direct and simple approach that correlated the annual oil production from olive trees with the accumulated absorbed PAR (they assumed that 0.12 g of olive oil are produced per 1 MJ of PAR).

Riaz et al. [96] modelled two APV solutions, comparing south-oriented monofacial PV modules with east–west-oriented bifacial ones. The study correlated crop yield under APV systems to the reduced irradiation on ground compared to open-field, employing empirical fitting parameters from previous studies. Model validation were done using data from Marrou et al. [129] and Elamri et al. [130]. This study highlighted the importance of considering different PV module orientations, such as vertical East-West and tilted North-South configurations, to optimise both crop yield and energy conversion. Specifically, the vertical East-West bifacial orientation demonstrated more spatially uniform light distribution, which is particularly beneficial for monocrop farming, while the tilted North-South configuration showed potential advantages for polyculture farming by accommodating crops with varying shade tolerance. Amaducci et al. [22] developed an irradiation distribution model using Scilab® and integrated it with GECROS v3.0 to investigate the impact of shading on maize cultivation under a stilt-mounted dual-axis tracking APV system in Italy. With a one-hour time step, GECROS is particularly well-suited for capturing the high temporal resolution patterns of alternation between full light and shade under APV systems. Simulation revealed that the reduction in irradiation under the APV system, as compared to open-field conditions, resulted in lower mean soil temperature and evapotranspiration, while soil moisture was higher. The average simulated grain yield under the APV system exhibited both higher and more stable outcomes compared to open-field conditions. This underscores the potential of sophisticated mechanistic models to capture high-temporal-resolution patterns of light and shade.

Cuppari et al. [117] took a different approach by developing a model to analyse the financial risks associated with APV systems, using regression models based on climatic and agronomic variables. The application of this model extended to four distinct cases across two different states in the United States of America, focusing on three crops: alfalfa, soybean, and strawberries. Crop yield modelling involved the utilisation of statistical data, with regression models constructed based on multiple explanatory variables including maximum and minimum temperatures, average temperatures, GHI, and precipitation, among others. Their work demonstrated the versatility of crop models in

financial analysis, extending the application of APV studies beyond purely agronomic outcomes. Similarly, Tahir and Butt [33] did not incorporate a mechanistic crop model into their modelling framework. Instead, they assumed that crop yield is intricately linked to the ratio of useful PAR (i.e., PAR effectively contributing to photosynthesis) in APV settings compared to the useful PAR in open-field conditions.

Fernandez et al. [112] developed a dual APV model to evaluate the integrated performance of PV systems and crops within agricultural greenhouses. The model underwent application across four diverse locations, namely El Ejido in Spain, Pachino in Italy, Antalya in Turkey, and Vicente Guerrero in Mexico, encompassing a total of 15 different crops. Crop yield estimation involved combining incident spectral irradiance data with crop spectral absorption characteristics. The crop's response to light was modelled using a light response curve, specifically the relationship between PPFD and the photosynthetic rate. This was achieved through the fitting of a fourth-order polynomial equation to measured data. The PPFD values for plants were derived from measurements, and a correction for spectral mismatch was applied. The assimilated CO<sub>2</sub> over time served as a proxy for crop yield. In a related study, Katsikogiannis et al. [35] employed the light response curve methodology to estimate the impact of shading on highbush blueberry leaves. The light response curve, originally derived from Li et al. [151], was utilised to establish a correlation between reduced PAR resulting from various APV system configurations and the photosynthetic rate, serving as a proxy for crop yield.

Jing et al. [89] explored the impact of shading on lettuce in urban rooftop APV systems, using the denitrification–decomposition (DNDC) biogeochemical simulator developed by Li et al. [152]. This simulation tool requires inputs related to weather data (such as precipitation, maximum temperature, and minimum temperature), soil characteristics, and cropping details. A notable limitation of utilising the DNDC model for APV system simulation is its lack of dependence on solar irradiation as a climatic ecological driver, as highlighted by Li et al. [153]. Consequently, this model is likely to have limited applicability for APV system studies, where the simulation of irradiation and the subsequent reduction caused by PV module shadings is crucial.

Elamri et al. [130] underscored the necessity for a novel crop model capable of capturing the short-term fluctuations in irradiance induced by the shading of crops by PV modules. As a foundational framework, the authors utilised the existing Optirrig model and enhanced it by incorporating a routing mechanism that accounts for variations in stomatal conductance. They emphasised the pivotal role of stomatal conductance in characterising short-term water and irradiance stresses within the APV context. To improve the model, the authors refined the routine responsible for calculating plant transpiration by introducing a stress coefficient ( $K_s$ ). This coefficient was defined as the ratio of the sum of short-term variations in stomatal conductance to the daily average stomatal conductance under open-field conditions. The upgraded dynamic model for stomatal conductance exhibited superior fitting to dynamic measurements compared to the static model. The enhanced model yielded satisfactory results in crop yield modelling, with the slope of the regression line ranging between 0.94 and 1.05.

Campana et al. [23] devised a multi-objective optimisation model for a vertically mounted APV system. This involved the integration of a PV system model, a shading model accounting for both ground and PV module shading, and the EPIC crop model. The weather parameters driving the crop model were determined based on the microclimate influenced by the PV modules, particularly focusing on reduced PAR, decreased evapotranspiration, and increased soil water moisture. Two crops (oats and potatoes) were subjected to simulation. The EPIC model demonstrated satisfactory performance, particularly highlighted in the subsequent study by Campana et al. [132] for oats. Significantly, the work by Campana et al. [23] stands out as one of the pioneering studies in the literature where the WEF nexus of APV systems is not only simulated but also optimised.

Trommsdorff et al. [45] adopted a similar methodology to compute

biomass yield in an experimental overhead APV system situated in Germany. The approach was based on the assessment of the PAR available beneath the APV modules in comparison to open-field conditions. The extent of available PAR was contingent on the row-to-row distance of the PV modules. To estimate biomass yield, the authors developed fitting equations based on experimental data, categorised according to crop classifications as PLUS, ZERO, and MINUS, as defined by Beck et al. [154].

Marrou et al. [129] conducted an analysis of the impact of APV shading on crop water utilisation by employing a water balance approach and soil water measurements to calculate actual evapotranspiration. The authors combined an experimental approach with a theoretical framework that involved breaking down actual evapotranspiration into two primary components – crop transpiration and soil evaporation. Additionally, they considered four key biophysical drivers: weather conditions, crop ground coverage, stomatal conductance, and soil surface hydraulic conductance. In the case of lettuce, actual evapotranspiration in the APV experimental systems accounted for 76 % and 78 % of the evapotranspiration in open-field conditions for the full density (where 50 % of incident irradiation reached the crop) and half density (where 70 % of incident irradiation reached the crop) APV systems, respectively. For cucumber, the corresponding values were 86 % and 91 %, respectively. The variance in actual evapotranspiration reduction was primarily attributed to the greater soil coverage of lettuce, leading to a corresponding reduction in soil evaporation. The authors demonstrated that the primary driver for actual evapotranspiration was irradiation reduction, although this did not consistently increase water use efficiency.

It is evident that various methodologies and models have been developed to understand the impact of PV module shading on crop growth. The studies surveyed in this section highlight the importance of integrating crop models with irradiance models, considering PV module orientation, and developing more sophisticated and location-specific models. Table 8 summarises the key outcomes, challenges, and future directions in crop modelling for APV systems.

### 3.4.2. Validation of crop models

In APV systems, understanding the interaction between crops and partial shading by PV panels is crucial. Validation of crop models plays a significant role in enhancing the prediction of crop growth and yield under these unique conditions. Table 9 summarises the outcomes of various studies focused on the validation of crop models in APV systems, including the specific crops studied, the models used, the parameters modelled, the approaches to validation, and the results obtained. These studies validate their models against actual field measurements or established data to confirm the reliability of the models under varying APV configurations.

While these models are generally effective in simulating crop responses to varying light and shade conditions, challenges remain in accurately modelling the diffuse and direct components of solar irradiation under different shading scenarios. Validation studies in APV systems emphasise the critical need for local calibration to refine predictions and enhance model reliability. For example, the AVIrrig model was validated for lettuce on a 4 m elevated APV system in Montpellier, France, using actual measurements of stomatal conductance and fresh biomass, achieving good agreement for fresh mass across PV configurations, with slopes ranging between 0.94 and 1.05 and MAE between 7 g and 23 g [130]. However, some models, such as the STICS crop model for durum wheat, struggled with dense shade, showing limitations in simulating leaf temperature and leaf area index under shaded conditions [2]. Furthermore, crop-specific models like APSIM-Oryza for rice showed slight underestimation of grain yield and total biomass, with relative RMSE values of 11.95 % when tested on a 3.5 m elevated APV system in Chikusei, Japan [92]. For grass, the AgriOptiCE® model underestimated dry mass by approximately 7 % under shading conditions but improved accuracy when input parameters were

**Table 8**

Key outcomes, challenges, and future directions in crop modelling for APV systems.

| Key Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Challenges                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Future Directions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Yield reductions observed due to shading can be mitigated by selecting crops that are better suited for specific APV configurations.</li> <li>Successful integration of crop growth models with irradiance models to simulate the effects of shading.</li> <li>Importance of considering PV module orientation and density in crop yield optimisation.</li> <li>Hourly resolution crop model GECROS is well-suited for capturing the high temporal resolution patterns of alternation between full light and shade</li> </ul> | <ul style="list-style-type: none"> <li>Existing crop models often lack the ability to simulate short-term irradiance fluctuations accurately.</li> <li>Need for more sophisticated models to capture complex environmental interactions and physiological responses.</li> <li>High requirement for location-specific data for accurate model validation and calibration.</li> <li>Variability in empirical parameters makes it difficult to generalise findings across different regions and crops.</li> <li>Difficulty in scaling up from controlled experiments to real-world agricultural settings.</li> <li>Balancing the dual objectives of PV energy production and agricultural productivity remains complex.</li> </ul> | <ul style="list-style-type: none"> <li>Develop novel crop models capable of accurately capturing spatial and temporal variations in environmental factors due to PV shading.</li> <li>Enhance existing models to better simulate detailed physiological processes and interactions.</li> <li>Develop simpler crop models that still effectively capture essential relationships between shading and crop growth. Ensure these models are robust enough for practical applications while requiring less detailed input data.</li> <li>Expand field experiments to include a wider variety of crops and diverse APV configurations.</li> <li>Conduct long-term studies to assess the impacts of APV systems on soil health, crop quality, and overall farm sustainability.</li> <li>Integrate financial risk analyses into crop modelling to support decision-making for farmers.</li> </ul> |

calibrated [120]. These findings underscore the importance of tailoring crop models to local environmental conditions and integrating site-specific parameters such as shading patterns, microclimatic variations, and soil-water interactions. The diversity of crops studies and the variability in model performance indicate a need for further refinement to account for environmental complexities and to improve predictions of crop growth and yield under APV systems.

### 3.5. Optimisation and integrated platforms for agrivoltaic systems

In the domain of APV modelling, simulation, and optimisation, numerous studies have attempted to decrypt the relationships between APV system configurations and their impacts on crucial factors such as light distribution, crop yields, and electricity generation. However, while these investigations have shed light on various aspects of APV systems, there remains a notable gap in the incorporation of comprehensive optimisation approaches, also including economic aspects. In such context, integrated platforms for the simulation and optimisation of APV systems are code, software, or combination of software to perform a holistic modelling of APV systems including: shading on PV modules and on ground; effects of ground shading and PV structure on microclimate; and the effect of microclimate on water consumption, crop production, and energy conversion. The technical aspects concerning food, energy and water can then be used as a starting point for modelling economic, social, environmental, climate change, and land use competition aspects. Various frameworks and principles highlight the significance of holistic approaches to effectively address the WEF nexus in APV systems



**Table 9**

Summary of the studies addressing the validation of crop models in APV systems.

| Crop                               | Model                                | Modelled parameters                     | Validation approach                                                                                                                                                                                                                        | Validation results                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------|--------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lettuce [130]                      | AVIrrig (optimisation model)         | Stomatal conductance, fresh mass        | Validation on a 4 m elevated APV system against actual measurements of stomatal conductance and fresh biomass at Irstea Montpellier, France, during spring and summer, 2016                                                                | Stomatal conductance slightly underestimated (slope = 0.89, $R^2 = 0.35$ ); good agreement for fresh mass across PV configurations (slope between 0.94 and 1.05, MAE between 7 g and 23 g)                                                                                                                                                           |
| Durum wheat [2]                    | STICS crop model                     | Dry mass, crop yield                    | Validation on a 4 m elevated APV system in Montpellier, France, measuring different PV densities across subsystems from November 2012 to June 2013                                                                                         | Reliable for dry mass; limitations in simulating under dense shade; suggested errors in leaf temperature and leaf area index calculations under shade                                                                                                                                                                                                |
| Tomato [33]                        | Simplified crop modelling            | Crop yield                              | Experimental validation in Oregon, USA comparing different irrigation and shading conditions<br>Experimental crop yield obtained by Al-Agele et al. [78]                                                                                   | Observed trends closely matched model predictions; average difference of 0.04 in crop yield                                                                                                                                                                                                                                                          |
| Soybean [155]                      | GECROS + shading algorithms          | Grain yield                             | Validation against observed grain yields for a dual-axis APV system with varying shade levels in Monticelli D'Ongina, Italy, 2021                                                                                                          | Underestimation in shaded areas (nRMSE of 16.5 % and 15.7 %)                                                                                                                                                                                                                                                                                         |
| Rice [92]                          | APSIM-Oryza model                    | Grain yield, total biomass              | Validation against measurements on a 3.5 m elevated APV system in Chikusei, Japan, May to October 2018                                                                                                                                     | Slight underestimation of grain yield and total biomass (relative RMSE of 11.95 %)                                                                                                                                                                                                                                                                   |
| C3 perennial grassland [156]       | Sperry model                         | Plant water potential                   | Validation on a single-axis APV system across different plots in Colorado, USA, May to August 2022                                                                                                                                         | Good correspondence except under negative pressure conditions; RMSE varied between 0.92 and 1.37 MPa                                                                                                                                                                                                                                                 |
| Ley grass [120]                    | Analytical crop model in AgriOptiCE® | Dry mass                                | Validation on a vertical APV system with dry mass measurements from three different cuts in 2022 at Kärrobo Prästgård, Sweden                                                                                                              | The calibrated model had a difference of 19 % compared to the actual measurements of the average crop yield under the APV system. Supplying the model with adjusted input parameters enhanced the model accuracy. The developed model underestimated the crop yield under shading conditions by ~7 % compared to the actual average measured values. |
| Lettuce, cucumber [129]            | Analytical and conceptual framework  | Fraction of soil covered by the crop    | Validation on a 4 m elevated APV against calculated soil cover ratios from image analysis and direct PAR measurements in Montpellier, France, 2011                                                                                         | Accurate modelling of soil cover progression for both crops under varying PV densities                                                                                                                                                                                                                                                               |
| Soybean, strawberry, alfalfa [117] | Regression model                     | Crop yield                              | Validation on a ground-mounted APV system against historical crop yield data using county-level data in Oregon and North Carolina, US                                                                                                      | Regression models resulted in $R^2$ values between 0.47 and 0.69; could overestimate by up to 17.6 % or underestimate by up to -13.5 %.                                                                                                                                                                                                              |
| Lettuce [96]                       | Analytical model                     | Crop yield                              | Validation on a 4 m elevated APV system by using reported yield data by Elamri et al. [63], considering solar irradiance reduction effects                                                                                                 | Model slightly overestimated crop yield, highlighting the need to consider more environmental factors                                                                                                                                                                                                                                                |
| Corn [157]                         | APSIM (calibrated with local data)   | Crop yield                              | Validation on a single-axis APV system against experimental yield measurements in different rows of crops.                                                                                                                                 | The agreement between simulation and experimental values was within $\pm 1$ %, with a slight tendency of the model to underestimate the measured yields.                                                                                                                                                                                             |
| Tomato [99]                        | TOMGRO                               | Crop yield at the end of each month     | Validation on a greenhouse against experimental values from the study of Hemming et al. [158] and simulation values from the work of Gong et al. [159]. Comparison of final weights of mature tomatoes harvested at the end of each month. | The presented model underestimated both the experimental and the simulated values, with deviations of ~14 % at the end of the crop cycle.                                                                                                                                                                                                            |
| Apple [160]                        | Kinetic fruit model                  | Soluble sugar and starch concentrations | Validation on a 5 m elevated APV system against experimental quality measurements during three consecutive years (from 2019 to 2021) in three different APV configurations.                                                                | The dynamics of the simulations of both physiology parameters followed the same trend as the measurements. RMSE between the measured data and the simulations were < 0.8 % for soluble sugar and < 0.7 % for starch concentration in all years and configurations.                                                                                   |

[161]. APV integrated modelling and optimisation platforms present a unique opportunity to tackle multiple aspects of the WEF nexus simultaneously to ensure comprehensive and global optimal solutions. The APV nexus is illustrated by Fig. 16, highlighting its key elements of agriculture, photovoltaics, microclimate, water, soil, biodiversity and their interconnections.

To optimise APV system design before installation, or to evaluate the performance after installation, Key Performance Indicators (KPIs) are essential metrics to achieve specific objectives or to assess specific aspects of the APV operation. In the context of APV systems, KPIs serve as quantifiable measures to evaluate system cross-cutting performances such as crop yield and quality, land use efficiency, energy conversion, water consumption and use efficiency, and economic and financial viability. A comprehensive understanding of these KPIs is crucial for optimising APV systems for sustainable agricultural practices and energy conversion. Table 10 summarises common KPIs used for evaluating or optimising APV systems, along with their descriptions and corresponding equations.

One of the first attempts to develop integrated simulation tools for

APV systems was performed by Dupraz et al. [2] by combining a solar irradiance model and the STICS crop model. The integrated tool was deployed to study the effects of the shadings produced by APV with different PV modules densities on durum wheat. The Land Equivalent Ratio (LER) was used as a KPI. Techno-economic optimisation was not deployed. Dinesh and Pearce in 2016 [168] used the STICS crop model to investigate the effects on lettuce yields when grown in different APV configurations, specifically between rows of ground-mounted PV modules and underneath stilt-mounted PV modules. They used PVsyst® to analyse the respective PV power production in these configurations. The study found that the shading caused by the PV modules significantly affected lettuce yields, with different configurations (full density, half density, and full sun) resulting in varying levels of yield reduction. Amaducci et al. [22] integrated, in the Scilab® environment, an irradiance on-ground model and GECROS v3.0 to investigate the effects of shadings on maize grown under stilt-mounted dual-axis tracking APV systems in Italy. The electricity production from PV modules was assessed using a static efficiency after computing the irradiance on PV modules, and LER was adopted as a KPI to compare different APV



Fig. 16. A visual representation of the APV nexus.

configurations as compared to solely agriculture and reference optimised ground-mounted PV systems. APV systems were also compared to biogas systems highlighting that APV could achieve a significantly higher land productivity in terms of  $\text{kWh ha}^{-1}$ . Campana et al. [23] integrated the EPIC crop model in the open-source package OptiCE for simulating and optimising clean energy conversion systems. The integrated platform, Agri-OptiCE®, simulates irradiation on the ground, microclimate, crop yield, and PV production. The model was used to study the effects of shadings on oats and potatoes. The simulation platform was coupled with a genetic algorithm to optimise the system design based on three different objectives, namely LER, annual electricity production, and power fluctuation. The optimisation model applied to vertically-mounted APV systems in Sweden highlighted that LER is probably not the best KPI for APV systems' optimisation, since it might lead to unwanted system configurations that maximise LER at the expense of both crop and energy performances, as shown in Fig. 17. However, they did not consider the economic implications of their optimisation model.

Mengi et al. [46] introduced an optimisation methodology that integrated light and crop modelling to streamline the design process of APV systems. Their approach, leveraging multi-objective genomic optimisation, aimed at maximising both PV electricity generation and crop performance metrics. Despite the breadth of their methodology, they acknowledged limitations in accuracy, particularly in crop modelling and the ray-tracing algorithm employed for light assessment. The work carried out by Mengi et al. [46] was based on the work by Isied et al. [169] and by Zohdi [170] on greenhouses. Similarly, Riaz et al. [25] proposed a metric to evaluate solar irradiance sharing between PV arrays and crops, highlighting factors influencing light productivity. However, their study notably lacked the integration of optimisation techniques to enhance system efficiency. Furthermore, while studies like those by Willockx et al. [39] and Katsikogiannis et al. [35] provided valuable insights into the environmental and configuration aspects of APV systems, they did not fully address the crucial economic dimension and optimisation strategies. Riaz et al. [96] compared APV configurations and explored the influence of row distance and tilt angle on system

potential, without addressing economic feasibility or optimisation. Similarly, Elamri et al. [130] explored water-saving potential in APV systems but did not conduct an economic analysis or optimisation. Jing et al. [89] developed an integrated model to evaluate the feasibility of urban rooftop APV systems in Shenzhen, China by combining a GIS with biogeochemical and solar power simulations. Such approaches provide valuable insights for city planners, policymakers, and researchers regarding the deployment of APV systems in specific locations. Similarly, Elkadeem et al. [171] used GIS to estimate the amount of Swedish territory suitable for APV systems using a set of geospatial criteria and an ordinal priority approach based on a multi-criterion decision-making algorithm. However, GIS-oriented approaches have certain limitations and should not be considered a comprehensive solution. GIS approaches can provide valuable spatial information for large-scale assessment and guide preliminary decision-making processes, but they lack the capacity to decrypt the intricate interactions within the APV system and might lead to suboptimal decisions and designs of APV systems on a local level. More detailed models that incorporate high temporal and spatial resolution, specific PV modules, soil characteristics, and crop physiology can significantly enhance accuracy when estimating APV system performance in diverse environments. This provides a better understanding of APV system performance, enabling stakeholders to make informed decisions and maximise benefits. Additionally, detailed models can be fine-tuned to investigate the implementation of specific crops and APV designs, such as fixed-tilt [23,26] or tracking [22]. However, it is important to recognise that while detailed models offer increased accuracy, they come with higher computational demands and require more input parameters. As such, their implementation may be more suitable for research studies or specific projects, rather than for large-scale assessment, where a well-structured GIS-based approach can still provide valuable insights. Bellone et al. [172] developed a simulation-based decision support system for APV systems to compare the performances of four different APV system configurations (i.e., vertical, overhead, single-axis, and dual-axis) and design, in a range of environments across Italy. Instead of setting an actual optimisation algorithm, the authors combined a set of economic, agronomic, and technical KPIs in a multi-

**Table 10**

Key performance indicators for APV systems.

| KPI                                  | Description                                                                                                                                              | Equation                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Land Equivalent Ratio (LER)          | Evaluates the land use efficiency for APV systems compared to traditional PV installations and only agriculture [2,22,23,35,39,42,45,46,96,120,130,162]. | $LER = \frac{Y_{ag,APV}x}{Y_{ag,ref}} + \frac{Y_{e,APV}}{Y_{e,ref}}$ $Y_{ag,APV}$ = Crop yield in APV system. $x$ = Reduction factor for land loss near mounting structure. $Y_{ag,ref}$ = Crop yield in control area. $Y_{e,APV}$ = Energy conversion of APV system. $Y_{e,ref}$ = Energy conversion of conventional ground-mounted PV system. |
| Light Productivity Factor (LPF)      | Evaluate the efficiency of solar irradiance sharing for a particular crop and APV system configuration [25,42].                                          | $LPF = \frac{Y_{e,APV}}{Y_{e,ref}} + \frac{PAR_{u,APV}}{PAR_{u,ref}}$ $Y_{e,APV}$ = Electrical yield of APV system. $Y_{e,ref}$ = Electrical yield of conventional ground-mounted PV system. $PAR_{u,APV}$ = PAR at the crop level. $PAR_{u,ref}$ = PAR for open field.                                                                         |
| Monetary Equivalent Ratio (MER)      | Assesses the economic advantage of intercropping systems [163].                                                                                          | $MER = \sum_i \frac{r_i Y_{int}}{R_{max}}$ $r_i$ = Return obtained by multiplying unit market price and yield of crop $i$ . $Y_{int}$ = Total yield of intercrop. $R_{max}$ = Maximum return achievable from one of the sole crops.                                                                                                             |
| Water Consumption (WC)               | Measures the volume of water consumed per unit area of crop [164].                                                                                       | $WC = \frac{TWU}{A}$ $TWU$ = Total water use. $A$ = Total cultivated area under APV system.                                                                                                                                                                                                                                                     |
| Water Productivity (WP)              | Measures the fresh biomass yield of a crop relative to the total amount of water used, including both rainfall and irrigation [130].                     | $WP = \frac{Y_{ag,APV}}{TWU}$ $Y_{ag,APV}$ = Crop yield in APV system. $TWU$ = Total water use.                                                                                                                                                                                                                                                 |
| Ground Coverage Ratio (GCR)          | The ratio of area occupied by PV modules to total APV system area [165].                                                                                 | $GCR = \frac{A_{PV}}{A}$ $A_{PV}$ = PV modules surface. $A_{ground}$ = Total cultivated area under APV system.                                                                                                                                                                                                                                  |
| Light Homogeneity Index (LHI)        | Measures the uniformity of PAR distribution within the APV area [44,81].                                                                                 | $LHI = 100 \cdot \left( 1 - \frac{\sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}}{\bar{x}} \right)$ $x_i$ = Total PAR that reaches the ground level in each separate area. $\bar{x}$ = Average annual PAR. $n$ = Total number of separate areas.                                                                                             |
| Levelised Cost of Electricity (LCOE) | Indicates the cost of electricity conversion [39,162,166].                                                                                               | $LCOE = \frac{CAPEX + \sum_{t=1}^n \frac{OPEX_t}{(1+WACC)^t}}{\sum_{t=1}^n \frac{Y_{e,APV}(1-d)^t}{(1+WACC)^t}}$ $CAPEX$ = Capital expenditures. $OPEX$ = Annual Operation and maintenance expenses. $d$ = Annual system degradation. $n$ = Lifetime of PV modules. $WACC$ = Weighted Average Cost of Capital.                                  |
| Light Use Efficiency (LUE)           | Relates total agricultural yield to the total amount of irradiance reaching the crop [46].                                                               | $LUE = \frac{Y_{ag,APV}}{TLI}$ $Y_{ag,APV}$ = Crop yield in APV system. $TLI$ = Total amount of irradiance reaching the crop.                                                                                                                                                                                                                   |
| Net Present Value (NPV)              | Determines a project's profitability by considering cash inflows and outflows [120,166].                                                                 | $NPV = -ICC + \sum_{t=1}^n \frac{(CF_{in,t} - CF_{out,t})}{(1+WACC)^t}$ $ICC$ = Initial capital cost. $CF_{in,t}$ = Cash inflow at year $t$ . $CF_{out,t}$ = Cash outflow at year $t$ . $WACC$ = Weighted average cost of capital. $n$ = Lifetime period of the system.                                                                         |
| Power Conversion Efficiency (PCE)    | Measures the performance of PV modules [167].                                                                                                            | $PCE = \frac{P_{max}}{I_T \times A_{PV}}$ $P_{max}$ = PV module maximum power point. $I_T$ = Total incident irradiance. $A_{PV}$ = PV modules surface.                                                                                                                                                                                          |

criterion decision analysis approach to identify the best APV system configuration and design to grow processing tomatoes. Another significant development in integrated APV modelling platforms is the Python agrivoltaic simulation environment (PASE 1.0) [83]. This open-source framework offers flexibility to simulate both PV energy yield and crop growth, allowing users to select from various crop and PV models based on their specific requirements. For crop modelling, PASE provides options such as the SIMPLE model for basic parameterisation, the Gras-Sim model for grassland growth, and the widely used STICS model through co-simulation with JavaStics. These models provide different levels of complexity, enabling simulations for both simple and advanced agro-nomic studies. On the PV side, PASE employs a physics-based cell temperature model combined with options for irradiance calculations, including the Hay Davies Klucher Reindl (HDKR) model and ray casting algorithm. The latter offers high spatial and temporal accuracy in modelling light distribution, which is crucial for analysing shading impacts in APV systems. Users can also input custom weather datasets or access built-in integrations with global databases such as PVGIS,

ensuring adaptability to diverse locations and climates. PASE's modular architecture facilitates user-defined configurations of PV arrays, including fixed-tilt, vertical, or tracking systems. Validation studies have shown that PASE achieves high accuracy in predicting daily irradiance, with relative errors as low as 1 %, and has demonstrated its capability to model the land productivity of APV systems in multi-year analyses. The platform's ability to integrate light-sharing, crop modelling, and energy yield within a unified framework underscores its potential as a robust tool for optimising APV designs and supporting decision-making for researchers, engineers, and policymakers.

In summary, integrated modelling platforms for APV systems provide essential tools for bridging the complex interactions between agricultural production, energy conversion, and environmental factors. These platforms, ranging from the early efforts of coupling crop and PV models to the advanced, modular frameworks like AgriOptiCE® and PASE 1.0, highlights the evolution of APV modelling towards higher flexibility, accuracy, and usability. By integrating diverse aspects such as shading, microclimate effects, and multi-objective optimisation, these



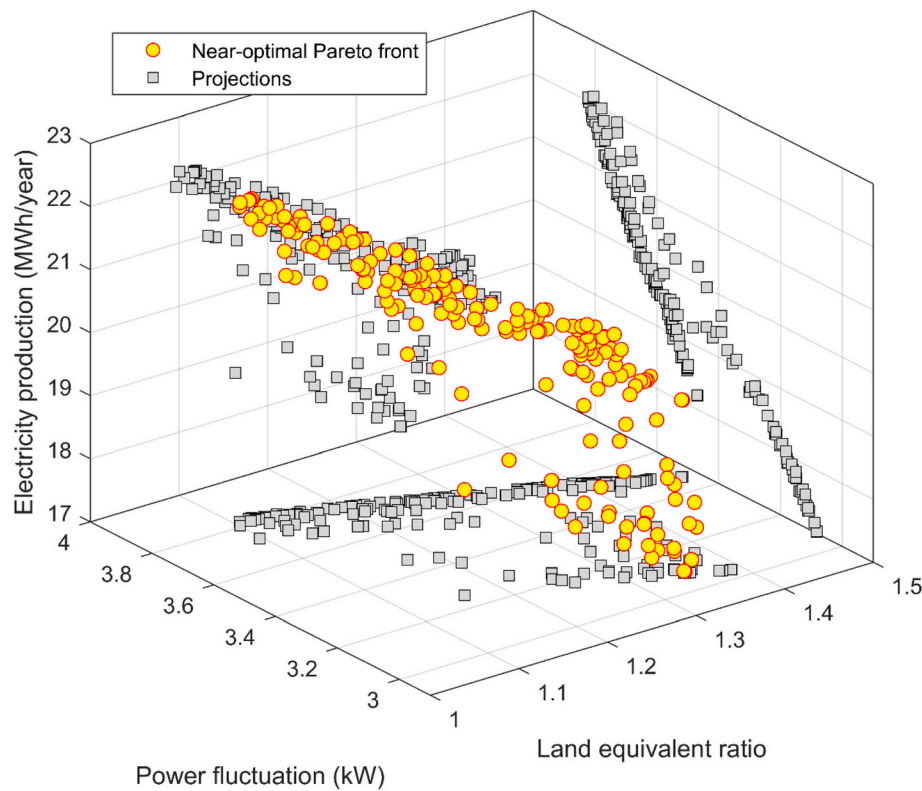


Fig. 17. Near-optimal Pareto frontiers of the three objectives optimisation and relative projections on the surfaces for vertically mounted APV systems [23].

tools pave the way for designing efficient and sustainable APV systems. Future advancements should focus on incorporating economic dimensions and further improving user accessibility, ensuring that these platforms can support both large-scale assessments and site-specific optimisation to address the pressing challenges of the WEF nexus.

#### 4. Conclusions

This study highlights the significant progress made in APV system modelling, showcasing a range of methodologies for irradiance distribution, crop growth, PV energy conversion, and microclimate simulations. This study also aims to highlight the critical role of KPIs for the optimal design of APV systems to comply with recent legislation, guidelines, standards, and regulations. Key advantages include the ability to optimise APV system designs for both energy and crop productivity, with models such as ray-tracing achieving high accuracy in irradiance simulations and crop-specific models like STICS and APSIM-Oryza effectively predicting yield outcomes under partial shading. Integrated approaches, combining these elements, offer a holistic perspective, facilitating the optimisation of dual-use systems to enhance the WEF nexus. Despite these advancements, several limitations persist. Shading models, while accurate, are computationally intensive, particularly when using ray-tracing techniques, limiting their scalability for large or complex systems. Many crop growth models require extensive local calibration to achieve reliable results, reducing their generalisability. Microclimate models face challenges in integrating parameters like wind flow, humidity, and air temperature variations across different APV configurations. Additionally, discrepancies between model predictions and real-world outcomes often arise due to unaccounted structural shading, resource competition, and weather-induced variability.

In conclusion, future research in APV should address several critical gaps to improve model accuracy, applicability, and comparability. Currently, the APV modelling field comprises diverse models originating

from both the PV and agricultural sectors, with emerging studies introducing novel model integrations. While many models demonstrate reasonable accuracy, the absence of standardised benchmarks creates confusion when selecting appropriate models for specific APV systems, applications, and geographical locations. Most studies validate their models against measurements, but this approach does not provide clarity on which models perform best for predicting crop yield and electricity production under varying conditions. There is a pressing need for studies that benchmark different APV models, enabling a better understanding of the trade-offs between computational complexity, interpretability, and accuracy. Physics-based models, such as PV single-diode models, CFD for microclimate, and biophysical models for crop modelling, offer high accuracy and detailed insights but are computationally intensive and harder to interpret. This drawback reduces their implementation into optimisation models. On the other hand, empirical models, such as those employing a static efficiency for electricity yield or simple linear relationships between shading and crop yield, are easier to interpret and computationally efficient but often lack precision (Fig. 18).

Hybrid models, which combine the strengths of both approaches, have started gaining attention. For instance, machine learning techniques can be used to simplify complex CFD simulations by serving as surrogate models or augmenting them with reduced computational costs. Such combinations allow researchers to retain accuracy while making the models more accessible and faster to run [173]. In scenario planning and potential analyses, simplified approaches such as empirical relationships between shading rate and crop yield reduction derived from meta-analyses offer a broader applicability but lower accuracy [174]. Conversely, case studies focusing on specific APV systems typically employ more detailed models to achieve accurate predictions of both crop and electricity yields. In such cases, further calibration is often applied to enhance models' accuracy, especially for crop models, but this introduces additional complexity, as it requires extensive data for each location and APV system [120]. This contrast underscores the need

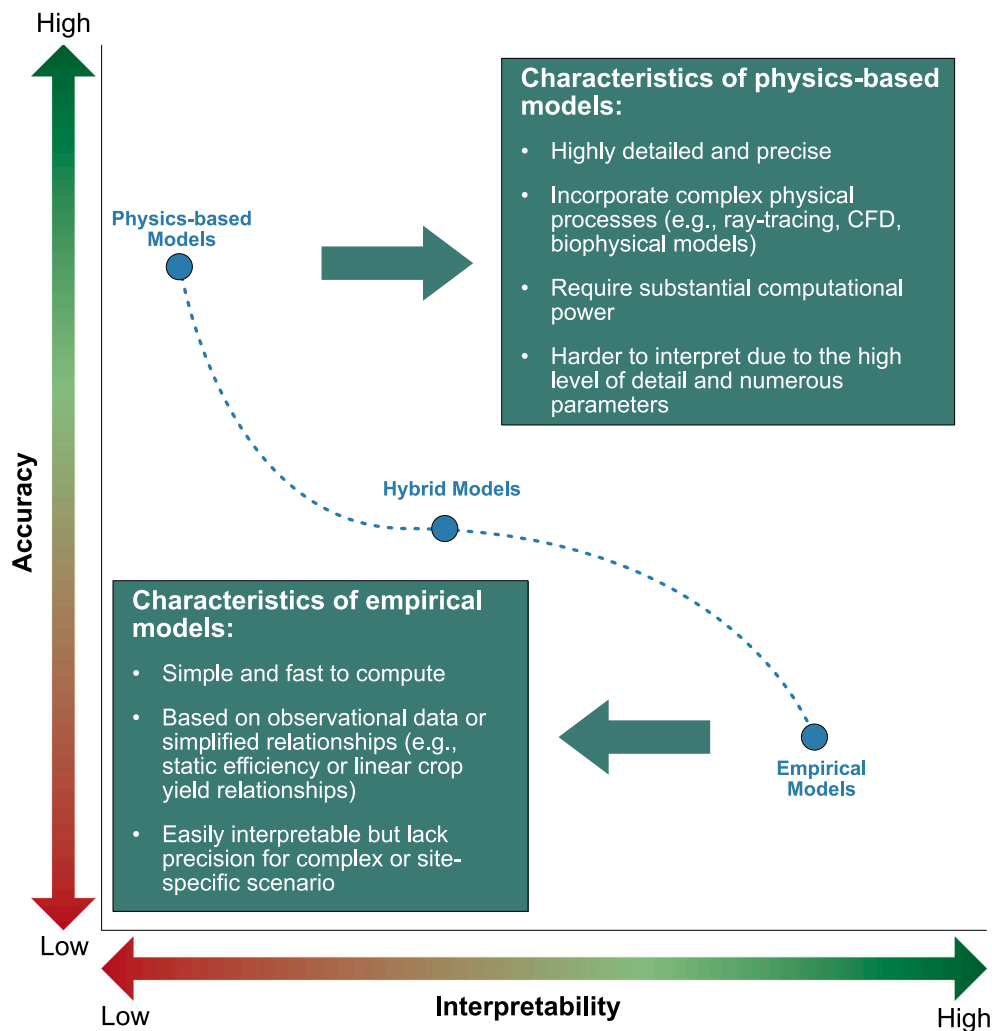


Fig. 18. Trade-offs between accuracy and interpretability in APV modelling approaches.

for a standardised baseline model to serve as a reference for evaluating new integrated APV models. For instance, in other research fields, such as the machine learning field, models benchmarks have been widely adopted, where standardised datasets and baseline models allow for consistent comparisons across algorithms [175–177]. These benchmarks provide a common ground for evaluating performance, enabling researchers to showcase improvements, validate new approaches, and demonstrate their models' advantages over existing ones. Adopting a similar approach in APV research would enhance comparability, as all new models would be assessed against the same benchmark, balancing computationally complexity and accuracy. This approach would reduce confusion and provide clear guidance for researchers in selecting the most appropriate modelling strategies. Future research should move beyond merely integrating crop and PV models and validating them against measurements. In literature, there are already available various shading, crop, microclimate, and PV models, whom combination could lead to several research papers with limited novelty and impact, but most likely all showing very accurate results after calibration. Moreover, most existing APV modelling efforts focus on conventional crystalline silicon (c-Si) technologies. With the growing interest in semi-transparent and wavelength-selective PV, such as organic and perovskites, to optimise light sharing for APV applications [178], future modelling frameworks will need to account for spectral composition and its impact on both energy conversion and plant growth. Additionally, there is a need for publicly available data, and collaborative efforts to ensure scalability and applicability across various APV configurations, crops, and regions.

For example, a recent study by Ma Lu et al. [179] has published data on crop yields under APV systems, providing a foundation for further validation and refinement of APV models. However, expanding such datasets across diverse climates, soil conditions, and PV configurations remains essential to enhance the accuracy and generalisability of future APV simulations.

#### CRediT authorship contribution statement

**Sebastian Zainali:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Silvia Ma Lu:** Writing – original draft, Writing – review & editing. **Álvaro Fernández-Solas:** Writing – review & editing, Writing – original draft. **Alejandro Cruz-Escabias:** Writing – review & editing, Writing – original draft. **Eduardo F. Fernández:** Writing – review & editing, Writing – original draft. **Tekai Eddine Khalil Zidane:** Writing – original draft. **Erlend Hustad Honningdalsnes:** Writing – review & editing, Writing – original draft. **Magnus Moe Nygård:** Writing – review & editing, Writing – original draft. **Jonathan Leloux:** Writing – review & editing. **Matthew Berwind:** Writing – review & editing, Writing – original draft. **Max Trommsdorff:** Writing – review & editing. **Stefano Amaducci:** Writing – review & editing, Writing – original draft. **Shiva Gorjian:** Writing – review & editing. **Pietro Elia Campana:** Conceptualization, Funding acquisition, Supervision, Writing – original draft, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

No data was used for the research described in the article.

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